The Symmetry-Quotient Attack on Odd-Characteristic SNOVA

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Abstract

The proposed odd-characteristic ($q = 19$) SNOVA Version 2.3 construction loses quadratic dimension for every symmetric public key. A base form labels features by ordered power pairs (ξ, η), but symmetry makes the polynomials labeled by (ξ, η) and (η, ξ) identical. If $A = \mathbb{F}_q[S]$ has dimension d, the d^2 labels factor through $\mathrm{Sym}^2(A)$ and span at most $\binom{d+1}{2}$ dimensions: 16 becomes 10 for $\ell = 4$, and 4 becomes 3 for $\ell = 2$. An exact public output split turns this identity into complete square forgery systems for all nine published parameter shapes, conditional on explicit public rank preflights, while retaining every verifier equation and the original rejection rule.

Finite-cardinality symbolic homotopy gives selected-degree-free recovery for all nine systems. In a unit-leading multiplication ledger, before an unmeasured multiplier κ_{hom}, the worst adaptive exponents are 132.188, 183.795, and 233.844. These are break-even operation counts, not complete classical-gate bounds. We therefore add a separate explicit Boolean-ledger analysis of the Level-I $\ell = 2$ row. It applies the published *Just Guess* transformation to the 16 extension-field diagonal equations, streams assignments to eight remaining extension variables, and filters the result through the cross channel and the original verifier. The diagonal and cross coefficients are not independent after the native byte-reduced coefficients are transformed to Frobenius channels. We instead prove, by an exact finite enumeration of the official $\ell = 2$ block transform, that every nonzero conditional cross-channel functional has maximum atom at most 49/829. Every field operation in the published expected-operation ledger is assigned a Boolean-circuit cost, with declared reserves for control, filtering, and verification. Under the same random-system regularity premise used by the published Just Guess analysis, the per-trial ledger is below $2^{132.047}$ gates. If a trial supplies at least $19^{16}/4$ sign-canonical diagonal roots, the conditional cross-channel second moment gives success probability above 0.0961 and a success-adjusted exponent below $135.426 + \log_2 \kappa_{\mathrm{JG}}$. Here κ_{JG} explicitly includes candidate-generation regularity, transformation restarts, and any cost beyond the declared filtering reserve. The break-even condition is $\log_2 \kappa_{\mathrm{JG}} < 7.575$; a factor-16 sensitivity point gives $2^{139.426}$. This replaces the opaque homotopy multiplier on the numerically weakest row by a substantially narrower, transcript-checkable premise, but it is not presented as an unconditional complete-gate break.

The symmetry quotient, affine reduction, target split, and reconstruction are exact. Key-density and candidate-dispersion statements are separated from those identities and are proved in the code-faithful random-XOF model or replaced by explicit public preflights. Keeping symmetry and the present coupling requires 45.00%–60.71% more outputs merely to restore the former feature cap and eliminate the guaranteed full-affine square.

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1 Introduction

A multivariate signature scheme publishes a polynomial map

$$P : \mathbb{F}_q^N \longrightarrow \mathbb{F}_q^M$$

of total degree at most two. Given a hash-derived target $t \in \mathbb{F}_q^M$, the algebraic core of a forgery is to find x with $P(x) = t$, while also passing any format and rejection checks. Equation and variable counts are only a first approximation to the difficulty: public dependencies among quadratic features can expose linear constraints, remove entire output directions, or split the remaining system into algebraically useful blocks.

SNOVA is one of the nine digital-signature candidates that NIST advanced to the third round of its additional post-quantum signature process in May 2026 [1, 2]. The characteristic-two construction was subsequently attacked through its wedge structure [3, 4]. The public odd-characteristic Version 2.3 design studied here is a proposed response: it moves to $q = 19$ and symmetrizes the scalar-expanded public base matrices [5]. Our target is exactly that public construction, not the earlier $q = 16$ submission and not an unspecified future revision.

The hidden duplication. Let P_i be a symmetric public base matrix, let $A = \mathbb{F}_q[S]$, and write $\Sigma_\xi = I_n \otimes \xi$ for $\xi \in A$. The quadratic feature with left label ξ and right label η is

$$q_{i,\xi,\eta}(u) = u^\mathsf{T} \Sigma_\xi P_i \Sigma_\eta u.$$

Because the matrices are symmetric and the output is a scalar,

$$q_{i,\xi,\eta}(u) = q_{i,\eta,\xi}(u). \tag{1}$$

The verifier assigns two ordered names to the same polynomial. Thus the d^2 ordered labels per base form factor through the symmetric square and leave at most

$$\dim_{\mathbb{F}_q} \mathrm{Sym}^2(A) = \binom{d+1}{2}$$

independent features. In the published rows, $d = \ell$, so 16 labels become 10 for $\ell = 4$ and 4 become 3 for $\ell = 2$. This identification occurs before the public outer map. Later mixing cannot reconstruct the alternating directions that have already vanished.

From an identity to a complete forgery system. Beullens showed that an attacker can impose an affine relation among the signature columns [6]. We apply the symmetry quotient inside that framework and then split the output coordinates exactly. Output directions unreachable by the quotient quadratics become ordinary affine or target-consistency equations. The remaining coordinates form a canonical K-equation quadratic map. Every transformation is public, invertible on its image, and retains every verifier equation. A recovered root is mapped back to a full signature candidate and checked by the unmodified verifier.

The complete data flow is

$$\text{public verifier} \longrightarrow \text{affine column relation} \longrightarrow \text{symmetry quotient},$$

$$\text{exact output split} \longrightarrow \text{square residual system} \longrightarrow \text{reconstructed, verifier-checked signature}.$$

The nonlinear solver can fail and restart, but it can never make the wrapper accept an invalid signature.

The direct all-nine theorem. After the exact public rank preflights, every published parameter row leaves all K quotient equations on a public K-dimensional slice. Solving this complete square removes the logical premises that complicated earlier versions of the analysis: no selected Macaulay degree, exact-corank prediction, preferred Jacobian core, or family-count enumeration is needed. The direct theorem gives the following unit-leading AXN multiplication-ledger exponents. They are useful break-even values, not complete gate counts.

Table 1: Conditional, selected-degree-free attack ledgers for all nine parameter shapes. The $\ell = 4$ entries condition on the exact public structural preflight; all entries normalize the stated spectrum and solver-restart events. Values are base-two unit-leading AXN pointwise-multiplication-schedule exponents, excluding κ_{hom}.

Level	Parameters (v, o, ℓ, r)	Direct square	Adaptive	Reference
I	$(28, 5, 4, 4)$	142.607	128.246	143
I	$(48, 16, 2, 2)$	132.971	132.188	143
I	$(28, 4, 4, 5)$	142.606	128.246	143
III	$(40, 7, 4, 4)$	185.451	171.617	207
III	$(72, 24, 2, 2)$	183.795	183.795	207
III	$(38, 5, 4, 5)$	185.450	171.616	207
V	$(50, 9, 4, 4)$	227.560	214.558	272
V	$(96, 32, 2, 2)$	233.844	233.844	272
V	$(52, 6, 4, 6)$	227.559	214.557	272

The levelwise direct-square maxima are 142.607, 185.451, and 233.844. Exact public leading-Jacobian tests optionally select smaller cores; failure invokes the complete square. Mixing the branches before taking logarithms gives levelwise adaptive maxima

$$\boxed{132.188, \qquad 183.795, \qquad 233.844}.$$

These have nominal arithmetic headroom 10.812, 23.205, and 38.156 bits below NIST's numerical references 143, 207, and 272 [7]. The exact break-even conditions for the adaptive maxima are

$$\log_2 \kappa_{\mathrm{hom}} < 10.812, \qquad 23.205, \qquad 38.156$$

at Levels I, III, and V. We do not assign a preferred value to κ_{hom}. The explicit Boolean-ledger Level-I route in Section 10 removes this multiplier from the numerically weakest row.

Why useful roots exist. For a slice direction y, let T_y be the polar derivative on the complete pre-elimination domain U, and define

$$\bar{\eta}_L^U = \sum_{0 \neq y \in L} 19^{-\operatorname{rank} T_y}.$$

The root experiment averages over the affine right-hand side as well as the slice coset, so the full domain U is the correct domain. If the verifier accepts density α and $\bar{\eta}_L^U \leq \eta$, then an accepted root at which the full restricted Jacobian has rank h occurs with probability at least

$$19^{h-K} \frac{a^2}{a + \eta}, \qquad a = \alpha - \eta/18.$$

A reserved public vinegar line separates slice selection from fresh internal public coefficients. In the random-XOF idealization of the official byte expansion, the aggregate spectrum lies within 2^{-128} of the exact full-rank baseline except with explicitly bounded probability. For a fixed key, a finite projective rank histogram certifies the same quantity exactly.

What is exact, modeled, and unclaimed. The quotient, output split, target filtering, rejection handling, reconstruction, determinant preflights, and final verifier check are exact public computations. The key-density statements use the code-faithful random-XOF idealization; exact fixed-key certificates are an alternative. The symbolic-homotopy complexity is theorem-backed, but κ_{hom} contains soft-O factors, evaluation and interpolation, basis transforms, inversion, factorization, filtering, memory traffic, control, and implementation constants. The Just Guess section replaces that multiplier by a Boolean-ledger instantiation of the published expected-operation schedule and a separate κ_{JG} recording candidate-generation regularity, restarts, and excess processing beyond the declared reserve. We do not claim a production timing, a low-memory homotopy theorem, an unconditional all-nine complete-gate break, or polynomial-time recovery: the displayed Bézout and guessing quantities remain exponential.

Design consequence. A repair retaining symmetry cannot be a small rejection tweak. Under the current parameter coupling, simultaneously restoring the former ordered-label feature cap and eliminating the guaranteed full-affine square requires 45.00%–60.71% more verifier outputs before any security margin is added. Eliminating the zero-offset chosen-message square by dimensions alone gives the stronger but narrower 101.5625%–144.00% floor. These are necessary, not sufficient, repaired dimensions.

Contributions. The paper makes five principal contributions.

1. It proves the basis-independent symmetric-square factorization and the exact quadratic-image cap for every symmetric public key.
2. It gives a verifier-preserving affine and target-consistency split for all nine public Version 2.3 shapes, including constructive rejection handling.
3. It proves a full-domain root-spectrum theorem and an accepted full-Jacobian-root bound that require no independence between the verifier's rejection rule and the polynomial system.
4. It gives complete-square and adaptive certified-core Las Vegas recovery theorems and an all-nine break-even ledger, and it instantiates a separate explicit Boolean-ledger Level-I sensitivity using Just Guess and the quotient-channel split.
5. It proves combined structural repair floors and retains a complementary streamed-XL branch for readers interested in explicit low-state arithmetic rather than the selected-degree-free headline.

Roadmap. Section 2 supplies the MQ, rank-spectrum, symbolic-homotopy, and underdetermined-MQ background. Section 3 states the public construction and symmetry quotient. Section 4 gives the exact affine-column and output reductions. Section 5 constructs fresh slices, handles rejection, and explains exact target sampling. The full-domain root and complete-square theorems begin in Section 6; the explicit Boolean-ledger Level-I route appears in Section 10. The low-state XL complement, implementation evidence, countermeasures, and conclusion follow afterward.

2 Background and relation to prior work

2.1 MQ systems, roots, and affine slices

An MQ system over $\mathbb{F}_q$ is a map $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$ whose coordinates have total degree at most two, together with a target $z \in \mathbb{F}_q^K$. Linear and constant terms are allowed. A root is an x satisfying $g(x) = z$. Underdetermination makes roots more plentiful but does not make the system linear or automatically easy.

An affine space is a translate $a + V$ of a linear subspace. Restricting the search to an affine slice $a + L \subseteq a + V$ turns the problem into a system in $h = \dim L$ coordinates. Three events must be separated:

(i) the target–slice pair contains a base-field root;

(ii) it contains an accepted root at which a chosen Jacobian has full rank;

(iii) the chosen algebraic solver recovers every such root within its stated work bound.

The first two are properties of the map and slice. The third is a solver statement. The paper proves them separately.

2.2 Polar forms, rank spectra, and collisions

For a scalar quadratic polynomial f, define the polar form

$$\beta_f(x, y) = f(x + y) - f(x) - f(y) + f(0).$$

In odd characteristic this is bilinear. For a vector-valued map g, a nonzero $b \in \mathbb{F}_q^K$ selects the scalar quadratic $b \cdot g$. Its polar rank measures how many independent directions the quadratic part couples.

For a fixed slice direction y, the vector-valued derivative

$$T_y(x) = g(x + y) - g(x) - g(y) + g(0)$$

is linear in x. The value $q^{-\operatorname{rank} T_y}$ is exactly the fraction of its codomain-dual directions that annihilate T_y, after the appropriate normalization. Summing these quantities over nonzero y controls the second factorial moment of the number of roots in a random target–coset pair. The same spectrum controls the density of points where the restricted Jacobian loses rank.

2.3 Macaulay matrices and the low-state branch

XL multiplies the input quadratics by selected monomials and treats the resulting polynomials as rows of a Macaulay matrix. Generic XL tracks total degree. Multigraded XL tracks one degree per variable block and can be much smaller when the system has a block structure [8, 9]. A one-dimensional right kernel is the easiest extraction case, but it is not the only sound one. If row reduction supplies a commuting border basis, the resulting multiplication matrices describe a finite quotient algebra; FGLM and finite-field factorization then enumerate its roots [10, 11]. The streamed-XL section later makes the additional production-degree premise fully explicit.

2.4 Finite-cardinality symbolic homotopy

The selected-degree-free branch uses a multihomogeneous homotopy theorem. We state the finite-field form used later.

Partition $N = \sum_{j=1}^{r} n_j$ variables into blocks $X_1, \dots, X_r$. Let $f_1, \dots, f_N$ be nonconstant polynomials represented by a straight-line program of length L, and suppose f_i has multidegree at most $\mathbf{d}_i = (d_{i,1}, \dots, d_{i,r})$. Define

$$B = C_{\mathbf{n}}(\mathbf{d}) = [\vartheta_1^{n_1} \cdots \vartheta_r^{n_r}] \prod_{i=1}^{N} \left(\sum_{j=1}^{r} d_{i,j} \vartheta_j \right), \tag{2}$$

$$B^+ = C'_{\mathbf{n}}(\mathbf{d}') = \operatorname{coeffsum} \left(\prod_{i=1}^{N} \left(\vartheta_0 + \sum_{j=1}^{r} d_{i,j} \vartheta_j \right) \bmod \langle \vartheta_0^2, \vartheta_1^{n_1+1}, \dots, \vartheta_r^{n_r+1} \rangle \right). \tag{3}$$

Theorem 2.1 (Finite-cardinality multihomogeneous homotopy)**.** *Let k be a perfect finite field and put*

$$E_0 = \max \left\{ \max_j \sum_i d_{i,j},\ 8B^2 \right\}.$$

If $|k| \geq E_0$, a randomized symbolic-homotopy algorithm whose separating form has a coefficient vector sampled uniformly from k^N returns a zero-dimensional parametrization containing every nonsingular root with probability at least $7/8$. On every run it returns a subset of the nonsingular roots or reports failure. Its arithmetic cost is

$$\widetilde{O} \left(BB^+ \left(L + \sum_{i,j} d_{i,j} + N^2 \right) N \right)$$

operations in k. This soft-O theorem does not instantiate its implied constant or polylogarithmic factors. More generally, if $|k| \geq \max_j \sum_i d_{i,j}$ and $|k| > B^2$, the same algorithm succeeds with probability at least $1 - B^2/|k|$.

Proof. Safey El Din and Schost prove the same homotopy construction under a large-characteristic hypothesis [12]. The two uses of that hypothesis can be replaced by finite-cardinality arguments.

For block j, put $D_j = \sum_i d_{i,j}$ and choose distinct $\alpha_{j,0}, \dots, \alpha_{j,D_j-1} \in k$. The affine moment-curve forms

$$\kappa_{j,u}(X_j) = X_{j,1} + \alpha_{j,u} X_{j,2} + \cdots + \alpha_{j,u}^{n_j-1} X_{j,n_j} + \alpha_{j,u}^{n_j}$$

have the same Vandermonde properties as the start forms in the cited proof: any n_j are independent and any $n_j + 1$ are inconsistent. The product start system can be written explicitly. Put

$$s_{i,j} = \sum_{\nu=1}^{i} d_{\nu,j}, \qquad s_{0,j} = 0,$$

and, for $1 \leq i \leq N$, set

$$g_i(X) = \prod_{j=1}^{b} \prod_{u=s_{i-1,j}}^{s_{i,j}-1} \kappa_{j,u}(X_j).$$

The factor ranges are disjoint. An isolated common zero chooses, for each i, one vanishing factor of g_i; exactly n_j choices must come from block j. The coefficient in (2) counts these choices, including the $d_{i,j}$ possible factors. The Vandermonde property gives one point for each choice, while inconsistency of any $n_j + 1$ forms prevents additional vanishing factors. The corresponding block-diagonal linear Jacobian is nonsingular, so all B roots are simple. They are enumerated within the same soft-linear cost. The first term in E_0 supplies enough distinct nodes.

The homotopy proof next requires a well-separating linear form. It identifies at most B^2 nonzero vectors on which the coefficient vector must not vanish. The cited proof samples from a one-parameter moment-curve family, which is why its field-size condition includes an additional factor depending on N. Here the coefficient vector is instead sampled uniformly from the full space k^N. For each fixed nonzero vector, its annihilator inside k^N is a proper k-linear subspace (possibly of codimension greater than one), so a uniform coefficient vector annihilates it with probability at most $|k|^{-1}$. Since $|k| \geq 8B^2$, a union bound gives success at least 7/8.

The remaining Newton lifting, specialization, squarefree cleaning, and zero-dimensional parametrization use only field operations and inversion at nonsingular points. Finite fields are perfect, so the rest of the cited correctness and arithmetic analysis is unchanged. Finally, direct substitution into the original system and a Jacobian-rank test remove every spurious or singular output; hence a run returns only nonsingular roots or reports failure. □

The theorem is a time theorem and does not promise small resident state. A parametrization can contain $O(B)$ field elements, but this output scale is dominated asymptotically by the displayed BB^+ term, which is typically $\Theta(B^2)$. All soft-O factors, fixed transforms, factorization, filtering, memory traffic, and implementation constants are retained later in the route- and parameter-dependent multiplier κ_{hom}.

2.5 Relation to prior attacks

Beullens rewrote the SNOVA verifier in ordered bilinear power-pair coordinates and derived affine-column forgery attacks from rank defects in the outer map [6]. Our quotient acts one layer earlier: public-matrix symmetry identifies the power-pair polynomials before the outer map sees them.

The qualitative risk of odd-characteristic symmetrization had been noted. A 2025 presentation by the SNOVA team used an alternating-form count, warned that a wedge-style variant might apply, and called for further cryptanalysis [13]. NIST later described the symmetric odd-characteristic construction as a proposed response rather than a stable design [2]. That warning did not give the exact $\mathrm{Sym}^2(A)$ affine-column factorization, complete residual systems, target and rejection analysis, or concrete all-nine recovery costs established here.

The characteristic-two wedge attack applies to the submitted $q = 2^k$ construction [3]; our theorem targets the symmetric $q = 19$ response. Other analyses reinterpret scalar-expanded SNOVA as UOV or extension-field UOV for key recovery [14, 15, 16]. Ran's characteristic-two attack instead uses exterior-algebra and wedge structure [17]. Jin et al. develop a reduced symmetric algebra for odd-characteristic UOV-family systems [18]; the present work is different in that it identifies the exact public $\mathrm{Sym}^2(A)$ feature quotient, combines it with SNOVA's affine-column/output split, and obtains direct forgery systems rather than hidden-subspace recovery. The complementary generic-MQ and partitioning literature includes Hashimoto's underdefined optimizer, Just Guess, generalized partitioning, and recent F4 engineering [19, 20, 21, 22].

2.6 Notation at a glance

Table 2: Recurring notation.

Symbol	Meaning
(v, o, ℓ, r)	Public parameter tuple: vinegar blocks, oil blocks, block height, and columns per block.
$n = v + o$	Number of matrix-valued signature blocks.
$M = or\ell$	Number of scalar verifier outputs.
$N_0 = n\ell$	Length of one stacked signature column after the common-column relation.
$A = \mathbb{F}_q[S]$	Public commutative matrix algebra; in Version 2.3, $A \cong \mathbb{F}_{q^\ell}$.
$d = \dim_{\mathbb{F}_q} A$	Algebra degree; $d = \ell$ for every official row.
$m_1 = \lceil or/\ell \rceil$	Number of public base-form groups.
$K = m_1\binom{\ell+1}{2}$	Maximum number of distinct quotient quadratics.
Q_R	Public matrix mapping quotient features to verifier outputs for relation R.
W_R, G_R	Left-kernel basis and public left inverse used for the exact output split.
$C_{R,v}$	Affine constraint matrix after projecting away the quadratic image.
L, s, h	Solver slice, its A-dimension when A-linear, and its base-field dimension.
$\bar{\eta}_L^U, \eta_{128}$	Full-domain directional collision mass and the instantiated certified allowance.
$\rho_{\text{byte}} = 14/256$	Maximum atom of one XOF byte reduced modulo 19.
$\rho_{\text{cond}} = 49/829$	Maximum conditional atom used by the Level-I cross-channel filter.
AXN	Unit-cost two-input AND/XOR/XNOR circuit model, with free wiring and fan-out.
κ_{hom}	Per-route, per-row ratio of complete end-to-end homotopy work to the unit-leading multiplication-schedule ledger.
κ_{JG}	Candidate-regularity, restart, and excess-processing multiplier excluded from the Just Guess Boolean ledger.

3 Public forms and the symmetry quotient

3.1 Parameters and public forms

The public parameter tuple is (v, o, ℓ, r). There are $n = v + o$ matrix-valued signature blocks, each with ℓ rows and r columns. Stacking one column of all blocks gives a vector of length $N_0 = n\ell$. The verifier has

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \qquad M = or\ell, \qquad N_0 = n\ell. \tag{4}$$

The ceiling is the convention used by the pinned Version 2.3 reference implementation [5].

Let $S \in \mathrm{Mat}_\ell(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial. Its powers generate

$$A = \mathbb{F}_q[S], \qquad \Sigma_\xi = I_n \otimes \xi \quad (\xi \in A).$$

For the public rows, $A \cong \mathbb{F}_{q^\ell}$ and $\dim_{\mathbb{F}_q} A = \ell$. The scalar-expanded base matrices $P_1, \ldots, P_{m_1} \in \mathrm{Mat}_{N_0}(\mathbb{F}_q)$ are symmetric. For $\xi, \eta \in A$, define

$$B_i^{\xi,\eta}(x, y) = x^\mathsf{T} \Sigma_\xi P_i \Sigma_\eta y. \tag{5}$$

Setting $x = y$ gives the quadratic power-pair feature.

Table 3: Dimensions of the nine public $q = 19$ Version 2.3 parameter rows.

Level	(v, o, ℓ, r)	m_1	M	K	N_0
I	$(28, 5, 4, 4)$	5	80	50	132
I	$(48, 16, 2, 2)$	16	64	48	128
I	$(28, 4, 4, 5)$	5	80	50	128
III	$(40, 7, 4, 4)$	7	112	70	188
III	$(72, 24, 2, 2)$	24	96	72	192
III	$(38, 5, 4, 5)$	7	100	70	172
V	$(50, 9, 4, 4)$	9	144	90	236
V	$(96, 32, 2, 2)$	32	128	96	256
V	$(52, 6, 4, 6)$	9	144	90	232

3.2 The basis-independent quotient

For one base form, the ordered label space is $A \otimes_{\mathbb{F}_q} A$. The following theorem identifies a universal kernel.

Theorem 3.1 (Symmetric-square factorization)**.** *Let $A = \mathbb{F}_q[S]$ have dimension d, and assume $S^\top = S$ and $P_i^\top = P_i$. For each i, the map*

$$A \otimes_{\mathbb{F}_q} A \longrightarrow \mathbb{F}_q[u]_2^{\mathrm{hom}}, \qquad \xi \otimes \eta \longmapsto u^\top \Sigma_\xi P_i \Sigma_\eta u$$

is invariant under $\xi \otimes \eta \mapsto \eta \otimes \xi$. It factors through

$$\mathrm{Sym}^2_{\mathbb{F}_q}(A) = (A \otimes_{\mathbb{F}_q} A)/\operatorname{span}\{\xi \otimes \eta - \eta \otimes \xi\}.$$

Consequently the direct sum over m_1 base forms has rank at most $m_1\binom{d+1}{2}$. After any public linear map to M outputs, its rank is at most

$$\min\left\{M,\ m_1\binom{d+1}{2}\right\}. \tag{6}$$

If the minimal polynomial of S has degree ℓ, then $d = \ell$ and the ordered feature vector is obtained by duplicating the off-diagonal coordinates of an unordered feature vector.

Proof. Every element of A is symmetric. Since the displayed value is a scalar,

$$u^\top \Sigma_\xi P_i \Sigma_\eta u = \left(u^\top \Sigma_\xi P_i \Sigma_\eta u\right)^\top = u^\top \Sigma_\eta P_i \Sigma_\xi u.$$

Thus every alternating label $\xi \otimes \eta - \eta \otimes \xi$ lies in the kernel. The symmetric square has dimension $\binom{d+1}{2}$, and a later linear map cannot enlarge its image. When powers of S form a basis, the coordinate statement is exactly the duplication of each off-diagonal unordered pair into its two ordered positions. $\square$

The theorem is an upper bound: extra public dependencies can only reduce the quadratic image further. Every concrete attack checks the exact public rank before solving. The outer map cannot repair the loss because it sees only the already-equal polynomial values. The factorization itself uses only transposition symmetry. Odd characteristic is needed later for the familiar decomposition into symmetric and alternating tensors and for the polar-form analysis. The submitted nonsymmetrized $q = 16$ construction is not covered.

3.3 Frobenius-distance normal form

The symmetric square has a canonical coordinate system that will be useful for the smaller adaptive cores. For $A = \mathbb{F}_{q^d}$ and $j > 0$, define

$$\phi_j([x \otimes y]_{\mathrm{sym}}) = xy^{q^j} + x^{q^j}y, \qquad \phi_0([x \otimes y]_{\mathrm{sym}}) = xy.$$

If d is even and $j = d/2$, the value lies in $\mathbb{F}_{q^{d/2}}$.

Theorem 3.2 (Frobenius-distance normal form)**.** *There is an $\mathbb{F}_q$-linear isomorphism*

$$\mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_{q^d}) \cong \begin{cases} A^{(d+1)/2}, & d \text{ odd}, \\ A^{d/2} \oplus \mathbb{F}_{q^{d/2}}, & d \text{ even}, \end{cases}$$

given by the maps ϕ_j for the distinct Frobenius distances. In particular,

$$\mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_{q^4}) \cong \mathbb{F}_{q^4} \oplus \mathbb{F}_{q^4} \oplus \mathbb{F}_{q^2}, \tag{7}$$

$$\mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_{q^2}) \cong \mathbb{F}_{q^2} \oplus \mathbb{F}_q. \tag{8}$$

Proof. Extend scalars to an algebraic closure and identify $A \otimes_{\mathbb{F}_q} \overline{\mathbb{F}}_q$ with a product of d copies indexed by $\mathbb{Z}/d\mathbb{Z}$. Frobenius cyclically shifts the primitive idempotents. The map ϕ_0 records the d diagonal unordered pairs. For $0 < j < d/2$, the conjugates of ϕ_j record the orbit of unordered pairs $\{i, i+j\}$, an orbit of size d. If d is even and $j = d/2$, the pair is repeated after a half-turn, leaving an orbit of size $d/2$. These orbits partition all unordered pairs. The displayed map is therefore a coordinate permutation after scalar extension and hence an isomorphism over $\mathbb{F}_q$. $\square$

For $\ell = 4$, the three channel types have highest homogeneous forms of ordinary degrees 2, $q + 1 = 20$, and $q^2 + 1 = 362$ over $A = \mathbb{F}_{19^4}$. For $\ell = 2$, the complete quotient becomes two eigenblocks with bidegree families $(2, 0)$, $(1, 1)$, and $(0, 2)$ after scalar extension.

4 The exact affine-column reduction

4.1 Imposing one common column variable

Write the stacked signature columns as

$$U = [u_0 \mid \cdots \mid u_{r-1}], \qquad u_j \in \mathbb{F}_q^{N_0}.$$

The attacker chooses multipliers $R_j \in A$ and offsets $v_j \in \mathbb{F}_q^{N_0}$, with $R_0 = 1$, and imposes

$$u_j = \Sigma_{R_j} u + v_j, \qquad 0 \le j < r. \tag{9}$$

This is a publicly checkable subset of possible signatures and makes no assumption about the honest signer.

Before substitution, the verifier feature vector has $N_{\mathrm{raw}} = m_1 \ell^2 r^2$ labeled coordinates. Let E be the public outer map after the fixed permutation that reconciles the official flattening order with the block-transposed order used here. For a fixed relation R, let R_R expand the remaining ordered power-pair features into the raw feature vector, and let $F_{R,v}$ collect the offset-linear coefficients. The complete substituted verifier has the form

$$E(R_R z_{\mathrm{ord}}(u) + F_{R,v} u) + c_{R,v} = t. \tag{10}$$

Put

$$E_R = ER_R, \qquad \Lambda_{R,v} = EF_{R,v}.$$

Products of two variable parts give the quadratics, products of one variable and one offset give the linear term, and products of two offsets give the constant.

Let $z_{\mathrm{sym}}(u)$ contain one coordinate for each unordered pair $0 \le a \le b < \ell$ in every base form, and let D duplicate its off-diagonal coordinates. By Theorem 3.1, $z_{\mathrm{ord}}(u) = Dz_{\mathrm{sym}}(u)$. Define

$$Q_R = E_R D \in \mathbb{F}_q^{M\times K}, \qquad K = m_1\binom{\ell+1}{2}. \tag{11}$$

The columns of Q_R span every output direction reachable by the quotient quadratics.

4.2 Canonical output coordinates

Choose W_R whose rows form a basis of the left kernel of Q_R, and put

$$C_{R,v} = W_R\Lambda_{R,v}. \tag{12}$$

Multiplication by W_R kills every quadratic feature and exposes ordinary affine constraints.

Lemma 4.1 (Canonical residual coordinates)**.** *Assume* $\operatorname{rank} Q_R = K$*. Choose a public left inverse* $G_R \in \mathbb{F}_q^{K\times M}$ *satisfying* $G_R Q_R = I_K$*. Then*

$$\begin{pmatrix} G_R \\ W_R \end{pmatrix} \in \mathbb{F}_q^{M\times M}$$

is invertible. Writing $b = t - c_{R,v}$*, the complete substituted verifier is equivalent to*

$$C_{R,v}u = W_R b, \qquad g_{R,v}(u) = G_R b, \qquad g_{R,v}(u) = z_{\mathrm{sym}}(u) + G_R\Lambda_{R,v}u. \tag{13}$$

The polar map of $g_{R,v}$ *is exactly that of* z_{sym}*. If* b *is uniform in* $\mathbb{F}_q^M$*, the two right-hand sides in* (13) *are independent and uniform.*

Proof. If a vector $y \in \mathbb{F}_q^M$ is killed by both G_R and W_R, then $W_R y = 0$ implies $y = Q_R a$ for some $a \in \mathbb{F}_q^K$. Applying G_R gives $a = G_R Q_R a = 0$, so $y = 0$. The stacked square matrix is therefore invertible. Applying its block rows to $Q_R z_{\mathrm{sym}}(u) + \Lambda_{R,v}u = b$ gives the two displayed systems, and invertibility gives the converse. Affine terms have zero polar. Finally, an invertible linear transformation sends a uniform vector to a uniform vector on the product of the two coordinate spaces. □

Proposition 4.2 (Exact affine reduction)**.** *Let* $\rho_Q = \operatorname{rank} Q_R$*,* $\lambda = \operatorname{rank} C_{R,v}$*, and* $\delta = M - \rho_Q - \lambda$*. Then*

$$\operatorname{rank}[Q_R\ \Lambda_{R,v}] = \rho_Q + \lambda.$$

After row reduction, exactly δ *independent affine consistency conditions remain on the target. For every target satisfying those conditions, Gaussian elimination leaves* ρ_Q *equations of degree at most two in* $N_0 - \lambda$ *variables, with exactly the same solution set as the complete substituted verifier.*

If $\rho_Q = K$ *and* $\lambda = M - K$*, every target passes the affine consistency test and the residual system consists of* K *equations on an affine translate of* $\ker C_{R,v}$*.*

Proof. The row space of W_R is $\ker Q_R^\mathsf{T}$, so $\ker W_R = (\ker Q_R^\mathsf{T})^\perp = \operatorname{im} Q_R$. Hence W_R realizes the quotient map $\mathbb{F}_q^M \to \mathbb{F}_q^M / \operatorname{im} Q_R$. The image of the columns of $\Lambda_{R,v}$ in this quotient has dimension $\operatorname{rank}(W_R \Lambda_{R,v}) = \lambda$, proving the first identity.

The projected affine system is $C_{R,v}u = W_R b$. Its codomain has dimension $M - \rho_Q$ and its image has dimension λ, leaving $\delta = (M - \rho_Q) - \lambda$ target consistency equations. Whenever they hold, parameterize the affine solution space by $N_0 - \lambda$ free variables and substitute into a complementary set of ρ_Q rows. Every operation is invertible row reduction or exact affine elimination, so the solution set is unchanged. In the full-rank case, $C_{R,v}$ is surjective and $\delta = 0$. $\square$

Operational meaning. The attacker computes $Q_R, G_R, W_R, C_{R,v}$ and the canonical residual map from the public key. Unfavorable public ranks cause a change of relation, offsets, or generated key before an expensive solve. Once the ranks pass, every solver root maps back through (9) to a candidate for the full original verifier. No equation has been dropped.

4.3 A Level-I dimension walkthrough

For $(v, o, \ell, r) = (28, 5, 4, 4)$, we have $n = 33$, $N_0 = 132$, $M = 80$, $m_1 = 5$, and $K = 50$. The common-column relation leaves 80 ordered quadratic labels, but the symmetry quotient leaves only 50 distinct quadratics. In the full-rank affine case, the other 30 output directions become linear equations and the full affine residual domain has dimension $132 - 30 = 102$. The complete-square theorem later chooses a fresh ordinary 50-dimensional slice inside that domain and solves all 50 residual equations together.

5 Stable slices, fresh coefficients, rejection, and targets

5.1 An unconditional stable kernel

Generic offsets can supply the missing affine output directions, but their linear terms may destroy the block grading used by multihomogeneous solvers. Choose one vector $w \in \mathbb{F}_q^{N_0}$ and multipliers $\Gamma_0, \ldots, \Gamma_{r-1} \in A$, and set

$$v_j = \Sigma_{\Gamma_j} w. \tag{14}$$

Define the offset row space

$$U_w = \operatorname{span}_{\mathbb{F}_q}\{w^\mathsf{T} \Sigma_\xi P_i \Sigma_\eta : 1 \le i \le m_1,\ \xi, \eta \in A\}. \tag{15}$$

For a basis $1, S, \ldots, S^{d-1}$ of A, form the complete right-A orbit

$$O_{R,v}^F = \begin{pmatrix} F_{R,v} \\ F_{R,v}\Sigma_S \\ \vdots \\ F_{R,v}\Sigma_{S^{d-1}} \end{pmatrix}, \qquad \rho_F = \operatorname{rank} O_{R,v}^F. \tag{16}$$

Theorem 5.1 (Stable-lift kernel)**.** *Assume $S^\mathsf{T} = S$, $P_i^\mathsf{T} = P_i$, and $A \cong \mathbb{F}_{q^d}$. Under (14):*

(i) U_w is closed under right multiplication by A and $\dim_{\mathbb{F}_q} U_w \le m_1 d^2$;

(ii) every row of $F_{R,v}$ and $O_{R,v}^F$ lies in U_w;

(iii) the row space of $O_{R,v}^F$ is an A-subspace, so $d \mid \rho_F$;

(iv)

$$H_{R,v}^F = \ker O_{R,v}^F$$

is A-linear, satisfies $F_{R,v}H_{R,v}^F = 0$, and has

$$\dim_A H_{R,v}^F = n - \rho_F/d \geq n - m_1 d. \tag{17}$$

No generic orbit-rank equality is required. An unexpected dependency only enlarges the stable space.

Proof. The spanning family in (15) has $m_1 d^2$ members. For $\zeta \in A$,

$$(w^{\mathsf{T}}\Sigma_\xi P_i \Sigma_\eta)\Sigma_\zeta = w^{\mathsf{T}}\Sigma_\xi P_i \Sigma_{\eta\zeta},$$

so U_w is right-A stable. Every offset cross term has one occurrence of w; if it appears on the right, transpose the scalar and use symmetry and commutativity to put it on the left. Thus all rows of $F_{R,v}$ lie in U_w, and right stability places all orbit rows there as well.

Because (16) contains the complete right orbit of every row, its row space is an A-vector space, proving $d \mid \rho_F$. Its first block is $F_{R,v}$, so the kernel kills every offset-linear feature. If x lies in the kernel and $\zeta \in A$, each product $S^a\zeta$ is an $\mathbb{F}_q$-linear combination of the chosen basis powers; therefore the same orbit rows kill $\Sigma_\zeta x$. The kernel is A-linear. Rank–nullity over A gives (17). □

Since $C_{R,v} = W_R E F_{R,v}$, the stable kernel lies inside $\ker C_{R,v}$. Translation along a subspace $L \subseteq H_{R,v}^F$ changes the quotient quadratics but leaves every offset-linear feature constant.

5.2 Eigenblocks and multidegrees

Let Ω split the minimal polynomial of S, and let $e_0, \dots, e_{\ell-1}$ be the primitive idempotents of $A \otimes_{\mathbb{F}_q} \Omega$. If L is A-linear of A-dimension s, then

$$L \otimes \Omega = L_0 \oplus \cdots \oplus L_{\ell-1}, \qquad \dim_\Omega L_a = s.$$

The unordered power-pair features are an invertible recombination of the unordered eigenblock features

$$x^{\mathsf{T}}\Sigma_{e_a} P_i \Sigma_{e_b} x, \qquad 0 \leq a \leq b < \ell.$$

They have multidegree $2\mathbf{e}_a$ when $a = b$ and $\mathbf{e}_a + \mathbf{e}_b$ when $a < b$. Translation creates lower-degree terms only in the same one or two blocks. One homogenizer per block therefore preserves these exact multidegrees. This is the grading used by the complementary streamed-XL branch.

Proof of the eigenblock statement. The power basis and the idempotent basis of $A \otimes \Omega \cong \Omega^\ell$ are related by an invertible Vandermonde matrix. Passing to the symmetric square gives an invertible change between unordered power-pair and unordered eigenblock coordinates. The idempotent e_a projects $L \otimes \Omega$ onto L_a. A diagonal feature uses one block twice; an off-diagonal feature is bilinear in two blocks. Translation and homogenization preserve the claimed supports. □

5.3 Reserved-line decoupling and the structural preflight

Let $e_1, \dots, e_n$ be the native A-coordinate basis in which the key-dependent symmetric public matrices are expanded. Fix a vinegar index j_0 and use the coordinate-aligned decomposition

$$W = Ae_{j_0}, \qquad V' = \bigoplus_{\substack{j \text{ vinegar} \\ j \neq j_0}} Ae_j, \qquad O_{\text{pub}} = \bigoplus_{j \text{ oil}} Ae_j,$$

$$V_{\text{pub}} = W \oplus V' \oplus O_{\text{pub}}, \quad \dim_A W = 1, \qquad \dim_A V' = v - 1, \qquad \dim_A O_{\text{pub}} = o. \tag{18}$$

The *native-coordinate random-XOF idealization* used below declares the native upper-triangular scalar entries of the key-dependent P_i to be independent reductions modulo 19 of independent XOF bytes. The deterministic outer map E, the relation R, and the complete multiplier vector $\Gamma = (\Gamma_0, \ldots, \Gamma_{r-1})$ are fixed and conditioned upon. No A-linear change of basis is applied before invoking independence. A reduced byte has maximum atom

$$\rho_{\text{byte}} = \frac{14}{256},$$

because nine residues have fourteen preimages and ten have thirteen.

Lemma 5.2 (Coordinate-aligned reserved-line decoupling). *Fix E, R, Γ, choose $0 \neq w \in W$, and use the structured offsets* (14)*. Then Q_R depends only on E and R, while $C_{R,v}$, $O^F_{R,v}$, $H^F_{R,v}$, and every deterministically tie-broken slice selected from them depend only on native entries having at least one index in W. In the native-coordinate random-XOF idealization they are independent of the native $V' \times V'$ entries.*

Proof. Every structured offset is in W. Each offset-linear coefficient therefore contains one copy of w; if it occurs on the right, transpose the scalar and use symmetry to put it on the left. Right multiplication by A preserves the first coordinate block. Thus neither the orbit construction nor any deterministic linear-algebra operation on it reads a native $V' \times V'$ entry. The two sets of raw XOF symbols are disjoint. Coordinate alignment is essential: the biased byte-reduction distribution is not invariant under an arbitrary A-linear basis change. □

Lemma 5.3 (Maximum-atom pivot bound). *Let $X_1, \ldots, X_N$ be independent finite-field variables, each with maximum atom at most ρ_{byte}. If a matrix B has rank t, then for every c,*

$$\Pr[BX = c] \leq \rho_{\text{byte}}^t.$$

Proof. Choose t pivot columns. Condition on all nonpivot variables. Expose the pivot variables in echelon order; at each step at most one value can satisfy the next independent equation, and that value has probability at most ρ_{byte}. Multiplication gives the claim. □

Put $\mathcal{Q}_R = \mathbb{F}_q^M / \operatorname{im} Q_R$. When $\operatorname{rank} Q_R = K$, $\dim_{\mathbb{F}_q} \mathcal{Q}_R = M - K$, and $C_{R,v}$ is a coordinate matrix for the induced linear map $u \mapsto W_R \Lambda_{R,v} u$ to $\mathcal{Q}_R$. Let $\mathbf{X}_{W,V'}$ be the vector of native independent public-matrix entries in the $W \times V'$ blocks. For $0 \neq \lambda \in \mathcal{Q}_R^\vee$, define B_λ by

$$\operatorname{vec}((\lambda \circ C_{R,v})|_{V'}) = B_\lambda \mathbf{X}_{W,V'} + b_\lambda.$$

Here B_λ and b_λ are fixed once E, R, Γ, λ and the native layout are fixed; they do not depend on the sampled entries in $\mathbf{X}_{W,V'}$. Scaling λ scales B_λ, so its rank depends only on $[\lambda]$. Put

$$t_{\text{src}} = \min_{0 \neq \lambda \in \mathcal{Q}_R^\vee} \operatorname{rank} B_\lambda.$$

For an A-line $\mathfrak{l} \subseteq W \oplus O_{\text{pub}}$, choose $0 \neq y_{\mathfrak{l}} \in \mathfrak{l}$ and write

$$O^F_{R,v} y_{\mathfrak{l}} = D_{\mathfrak{l}} \mathbf{X}_{\text{nat}} + e_{\mathfrak{l}}$$

in the complete vector $\mathbf{X}_{\text{nat}}$ of native independent public-matrix entries. The coefficient data $D_{\mathfrak{l}}, e_{\mathfrak{l}}$ are deterministic before those entries are sampled, and the rank of $D_{\mathfrak{l}}$ is independent of the chosen generator. Define

$$t_W = \operatorname{rank} D_W, \qquad t_{\text{oth}} = \min_{\substack{\mathfrak{l} \subseteq W \oplus O_{\text{pub}} \\ \dim_A \mathfrak{l}=1,\ \mathfrak{l} \neq W}} \operatorname{rank} D_{\mathfrak{l}}.$$

These are exact symbolic coefficient-map ranks determined by the fixed relation, the complete offset pattern, and the native coordinate layout; they do not depend on the sampled values of the key coefficients. Define

$$\delta_{\text{src}} = \frac{q^{M-K}-1}{q-1} \rho_{\text{byte}}^{d(v-1)}, \tag{19}$$

$$\delta_{\text{proj}} = \rho_{\text{byte}}^{m_1\binom{d+1}{2}} + \left(\frac{|A|^{1+o}-1}{|A|-1} - 1 \right) \rho_{\text{byte}}^{m_1 d^2}, \qquad |A| = q^d. \tag{20}$$

Theorem 5.4 (Conditional coefficient-preflight density)**.** *Fix the deterministic outer map E, a relation R, and the complete vector Γ. Assume the exact deterministic checks*

$$\operatorname{rank} Q_R = K, \qquad t_{\text{src}} \geq d(v-1), \qquad t_W \geq m_1 \binom{d+1}{2}, \qquad t_{\text{oth}} \geq m_1 d^2$$

pass. In the native-coordinate random-XOF idealization,

$$\Pr[\operatorname{rank} C_{R,v} < M-K] \leq \delta_{\text{src}},$$

and

$$\Pr[H^F_{R,v} \cap (W \oplus O_{\text{pub}}) \neq 0] \leq \delta_{\text{proj}}.$$

Consequently full affine rank and injectivity of $H^F_{R,v} \to V'$ hold jointly with probability at least $1 - \delta_{\text{src}} - \delta_{\text{proj}}$. For the six official $\ell = 4$ dimension tuples, direct substitution gives $\delta_{\text{src}} + \delta_{\text{proj}} < 2^{-209}$. The coefficient-map ranks are fixed-layout symbolic checks; the realized quotient, affine, and intersection ranks are exact public checks on a fixed key.

Proof. If $\operatorname{rank} C_{R,v} < M-K$, some nonzero $\lambda \in \mathcal{Q}_R^\vee$ annihilates $C_{R,v}$. In particular, $(\lambda \circ C_{R,v})|_{V'} = 0$. For fixed $[\lambda]$, this is an affine system of rank at least $d(v-1)$ in the independent native variables $\mathbf{X}_{W,V'}$; Theorem 5.3 bounds its probability by $\rho_{\text{byte}}^{d(v-1)}$. Scalar multiples define the same event, and $\mathbb{P}(\mathcal{Q}_R^\vee)(\mathbb{F}_q)$ has $(q^{M-K}-1)/(q-1)$ points. A union bound proves (19).

By Theorem 5.1, $H^F_{R,v}$ is A-linear. Thus a nonzero intersection with $W \oplus O_{\text{pub}}$ contains an A-line $\mathfrak{l}$. For $\mathfrak{l} = W$, the coefficient-rank hypothesis and Theorem 5.3 give probability at most $\rho_{\text{byte}}^{m_1\binom{d+1}{2}}$. For every other A-line the analogous bound is $\rho_{\text{byte}}^{m_1 d^2}$. The $(1+o)$-dimensional A-space $W \oplus O_{\text{pub}}$ contains $(|A|^{1+o}-1)/(|A|-1)$ lines, exactly one of which is W; a union bound gives (20). Finally, the kernel of the projection $H^F_{R,v} \to V'$ is precisely $H^F_{R,v} \cap (W \oplus O_{\text{pub}})$. A final union bound combines the two bad events without assuming independence. Direct substitution gives the displayed uniform numerical bound. □

No tail of Γ is implicit in this statement: the concrete route records all r entries and runs the displayed checks for the vector above. This release does not assert that it passes them for every fixed key. The six $\ell = 4$ ledgers below are conditional on a successful structural preflight.

5.4 The guaranteed fresh ordinary square

Let

$$D_0 = V' \cap \ker C_{R,v}.$$

The intersection inequality gives

$$\dim_{\mathbb{F}_{19}} D_0 \geq 4(v-1) - (M-K). \tag{21}$$

For all six $\ell = 4$ rows this lower bound is at least K:

Parameters	$\dim D_0$ lower bound	K
$(28, 5, 4, 4)$, $(28, 4, 4, 5)$	78	50
$(40, 7, 4, 4)$	114	70
$(38, 5, 4, 5)$	118	70
$(50, 9, 4, 4)$	142	90
$(52, 6, 4, 6)$	150	90

Consequently the attacker can choose, from cross data alone, an ordinary K-dimensional slice inside D_0. Its internal $V' \times V'$ coefficients remain fresh for the spectrum theorem. This simple dimension fact is the reason the complete-square route applies to every $\ell = 4$ row, including $(38, 5, 4, 5)$; it does not require the smaller structured stable kernel to have A-dimension $K/4$.

5.5 Passing the block-symmetry rejection rule

Version 2.3 rejects a square $\ell = 4$ signature if any block is symmetric and rejects an $\ell = 2$ signature if more than $\lfloor n/4 \rfloor$ blocks are symmetric [5]. Rectangular $\ell = 4$ blocks are not subject to the square-block rule.

For a square $\ell = 4$ relation $R = (R_0, R_1, R_2, R_3)$, define the local skew map

$$\mathcal{S}_R : \mathbb{F}_q^4 \longrightarrow \bigwedge^2 \mathbb{F}_q^4, \qquad \mathcal{S}_R(x)_{a,c} = (R_c x)_a - (R_a x)_c.$$

For any offsets, one signature block is symmetric exactly when $\mathcal{S}_R(x)$ equals a fixed offset-dependent vector. The public relation $R_j = S^j$ for the official matrix has $\operatorname{rank} \mathcal{S}_R = 4$. Thus each block has at most one rejected common-column value, and the accepted density in the full common-column space is at least

$$\alpha_{4,n} = (1 - 19^{-4})^n. \tag{22}$$

The accepted-root theorem later uses only this global density; it assumes no independence from the MQ system.

For $\ell = 2$, the zero-offset relation used in the main theorem gives a block

$$\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix},$$

which is symmetric exactly when $y = 0$. The exact accepted density is derived in (39). A deterministic alternative is also available: for the official matrix

$$S = \begin{pmatrix} 1 & 2 \\ 2 & 15 \end{pmatrix}, \qquad R_1 = 9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix},$$

choose offsets with $(v_1)_{b,0} - (v_0)_{b,1} = 1$ in every block. Then each block has fixed nonzero skew 1, so every root passes the rejection rule. This fixed-skew route is a useful independent check but is not needed for the selected-degree-free headline.

5.6 Exactly uniform official targets

The probability theorems use a uniform target over $\mathbb{F}_{19}^M$. The official binary-to-field conversion has a small modulo bias. The attacker removes it without changing the verifier. Here the target hash/XOF chunks are modeled as independent uniform bit strings, separately from the native-coordinate key-expansion model above. The pinned conversion uses each full eight-byte chunk for fifteen base-19 digits and a final partial chunk for the remaining digits. If a uniform b-bit chunk X is to supply a base-19 digits, retain the message–salt input only when

$$X < \left\lfloor \frac{2^b}{19^a} \right\rfloor 19^a. \tag{23}$$

Conditioned on this event, $X \bmod 19^a$ is uniform. Applying the test to each chunk makes the retained official target exactly uniform, and an invertible output change preserves uniformity. Exact evaluation over the nine public Version 2.3 lengths gives a minimum acceptance probability $0.1524622985\ldots$, and hence at least 0.152462. When target consistency requires more than the 128-bit salt space, the existential chosen-message attacker varies a message prefix as well as the salt. Target generation occurs before nonlinear solving and adds to, rather than multiplies, the solve cost.

5.7 The exact first and second moments

The accepted-root theorem rests on an exact collision count. We state it here so that the probability argument is self-contained.

Theorem 5.5 (Root-collision identity)**.** *Let U be an $\mathbb{F}_q$-space, let $C : U \to \mathbb{F}_q^c$ be surjective, let $L \subseteq \ker C$ have dimension h, and let $g : U \to \mathbb{F}_q^K$ have degree at most two. Sample independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, then choose a uniform affine coset $\mathcal{A}$ of L inside $C^{-1}(d)$. Put*

$$Z = |\{x \in \mathcal{A} : g(x) = z\}|, \qquad \mu = q^{h-K}.$$

For $0 \neq y \in L$, define

$$T_y(x) = g(x+y) - g(x) - g(y) + g(0)$$

and let $\kappa(y)$ be one if $-(g(y) - g(0)) \in \operatorname{im} T_y$ and zero otherwise. Then

$$\mathbb{E}Z = \mu, \tag{24}$$

$$\mathbb{E}[Z(Z-1)] = \mu \sum_{0 \neq y \in L} \kappa(y) q^{-\operatorname{rank} T_y} \leq \mu \sum_{0 \neq y \in L} q^{-\operatorname{rank} T_y}. \tag{25}$$

The domain of T_y is the full space U because the experiment averages over the affine right-hand side as well as the coset.

Proof. Sampling d and then a uniform L-coset in $C^{-1}(d)$ is exactly uniform over all cosets of L in U. Every point of U therefore lies in the sampled coset with probability $q^{h-\dim U}$, and its image matches the independent target with probability q^{-K}. Summing over all points gives (24).

For an ordered pair of distinct roots in one coset, write the second point as $x + y$ with $0 \neq y \in L$. The collision condition is

$$g(x+y) = g(x) \quad \Longleftrightarrow \quad T_y(x) = -(g(y) - g(0)).$$

It has $\kappa(y) q^{\dim U - \operatorname{rank} T_y}$ solutions in x. Divide the total number of ordered collisions by the number $q^{\dim U - h} q^K$ of target–coset pairs to obtain (25). □

6 Full-domain spectrum and complete-square recovery

The strengthened attack now has a deliberately simple primary theorem. After the exact symmetry quotient and affine split, every published row leaves a complete square residual system: K equations on a K-dimensional public slice. The full-domain spectrum theorem supplies accepted nonsingular roots, and finite-cardinality symbolic homotopy recovers them. No selected degree, chosen Jacobian core, family-count sweep, or exact-corank prediction appears in this direct route.

Layer	Mathematical statement	Status
Exact algebra	Public-matrix symmetry factors the ordered feature map through $\mathrm{Sym}^2(A)$; the output split retains every verifier equation and reconstructs every candidate.	Unconditional for every symmetric public key passing the stated exact public ranks.
Root density	A full-domain directional-rank spectrum bounds both collisions and the singular locus of the restricted Jacobian.	Theorem in the code-faithful random-XOF idealization, or an exact projective-rank certificate for a fixed key.
Complete-square recovery	Restrict the full K-coordinate residual map to a public K-dimensional slice and solve all K equations together. Every candidate is checked by the original verifier.	Selected-degree-free and fixed-core-free Las Vegas theorem with no orbit enumeration.
Certified optimization	An exact determinant preflight may replace the complete square by a smaller channel core; failure invokes the complete square or a lower-work orbit fallback.	A certified branch optimization, not an assumption about the public key.

The direct theorem is the conceptual headline. The adaptive cores improve its arithmetic margins, but they are not needed to establish a theorem-backed attack against any of the nine rows. This separation makes the logical claim independent of the optimization machinery.

The numerical statements below are unit-leading AXN operation ledgers. They are not wall-clock timings, low-memory theorems, or complete attack circuits. They make no polynomial-time claim: the displayed multihomogeneous Bézout quantities are exponential, and κ_{hom} remains visible.

6.1 The spectrum required by the accepted-root theorem

Let U be a finite-dimensional $\mathbb{F}_q$-space, let $C : U \to \mathbb{F}_q^c$ be surjective, let $g : U \to \mathbb{F}_q^K$ have degree at most two, and let $L \subseteq \ker C$ have dimension h. Write $g = q_2 + q_1 + q_0$ by homogeneous degree and define, for $0 \neq y \in L$,

$$T_y : U \longrightarrow \mathbb{F}_q^K, \qquad T_y(x) = g(x + y) - g(x) - g(y) + g(0), \tag{26}$$

$$\bar{\eta}_L^U = \sum_{0 \neq y \in L} q^{-\operatorname{rank} T_y}. \tag{27}$$

The superscript matters. The accepted-root experiment first samples a uniform affine right-hand side for C and then a uniform L-coset in that fiber. Its collision calculation averages over the whole domain U. Consequently the relevant derivative in (26) has domain U, not $\ker C$. Restricting T_y to $\ker C$ gives a different, generally larger quantity that is useful for fixed-fiber uniqueness but is not needed here.

Let $X \subseteq U$ be the assignments accepted by the original verifier and put $\alpha = |X|/|U|$. A trial samples independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, and then a uniform affine coset of L inside $C^{-1}(d)$.

Theorem 6.1 (Accepted nonsingular root, full-domain form)**.** *Assume $\bar{\eta}_L^U \le \eta$ and $\alpha > \eta/(q-1)$. A trial contains an accepted root x satisfying*

$$\operatorname{rank} D_L g(x) = h, \qquad D_L g(x)[y] = T_y(x) + q_1(y),$$

with probability at least

$$p_{\mathrm{ns}} = q^{h-K} \frac{a^2}{a+\eta}, \qquad a = \alpha - \frac{\eta}{q-1}. \tag{28}$$

No independence among acceptance, singularity, the target, and the quadratic map is assumed.

Proof. Let Y count accepted roots in the sampled target–coset pair at which $D_L g$ has rank h, and let Z count all roots. For a projective direction $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$, the condition $D_L g(x)[y] = 0$ is an affine system in x whose linear part is T_y. It therefore holds on at most a $q^{-\operatorname{rank} T_y}$ fraction of U. Union-bound over projective directions to obtain

$$\frac{|\{x : \operatorname{rank} D_L g(x) < h\}|}{|U|} \le \frac{1}{q-1} \sum_{0 \ne y \in L} q^{-\operatorname{rank} T_y} \le \frac{\eta}{q-1}.$$

Every point of U is selected by the random fiber–coset with probability $q^{h-\dim U}$ and matches the independent target with probability q^{-K}. Put $\mu = q^{h-K}$ and $a = \alpha - \eta/(q-1)$. Hence $\mathbb{E}Y \ge \mu a$. The ordered-pair collision identity gives $\mathbb{E}[Z(Z-1)] \le \mu\eta$, and $Y(Y-1) \le Z(Z-1)$. Therefore

$$\mathbb{E}[Y^2] = \mathbb{E}Y + \mathbb{E}[Y(Y-1)] \le \mathbb{E}Y + \mu\eta.$$

The function $x \mapsto x^2/(x+\mu\eta)$ is increasing for $x \ge 0$. Cauchy–Schwarz therefore gives

$$\Pr[Y > 0] \ge \frac{(\mathbb{E}Y)^2}{\mathbb{E}Y + \mu\eta} \ge \mu \frac{a^2}{a+\eta},$$

which proves (28). This strictly sharpens the former denominator $1+\eta$ whenever $a < 1$. $\square$

The following lemma improves the seeded theorem because it prices exactly the full-domain spectrum used above. Let $\rho_{\mathrm{byte}} = 14/256$ be the maximum atom of one public-expansion byte reduced modulo 19.

Lemma 6.2 (Full-domain fresh-spectrum theorem)**.** *Assume q is odd and $A = \mathbb{F}_{q^d}$. Let L be fixed from coefficient data disjoint from a fresh symmetric-matrix region. Suppose every $0 \ne y \in L$ has nonzero projection $\bar{y}$ to a fresh A-space D, and every nonzero residual-output combination contains a nonzero coefficient $c \in \mathrm{Sym}^2_{\mathbb{F}_q}(A)$ in at least one fresh public base form. If*

$$\dim_{\mathbb{F}_q} D = d + t,$$

then for all $0 \ne y \in L$ and $0 \ne b \in \mathbb{F}_q^K$,

$$\Pr[b \in \ker T_y^{\mathsf{T}}] \le \rho_{\mathrm{byte}}^t. \tag{29}$$

Consequently

$$\mathbb{E}[\bar{\eta}_L^U] \le (q^h - 1)q^{-K} + (q^h - 1)(1 - q^{-K})\rho_{\mathrm{byte}}^t, \tag{30}$$

$$\Pr\Big[\bar{\eta}_L^U > (q^h - 1)q^{-K} + \varepsilon\Big] \le \frac{(q^h - 1)(1 - q^{-K})\rho_{\mathrm{byte}}^t}{\varepsilon}. \tag{31}$$

For the reserved-line construction one may take $D = V' \cong A^{v-1}$ and $t = d(v-2)$. For the zero-offset construction on the full public-vinegar summand one may take $D \cong A^v$ and $t = d(v-1)$.

Proof. Fix $y \neq 0$ and $b \neq 0$, and choose a fresh public base form on which the corresponding symmetric-square coefficient c is nonzero. If $c = \sum_\nu a_\nu \odot b_\nu$, the coefficient vector of the fresh symmetric matrix in the scalar derivative $b \cdot T_y(x)$ is

$$\Xi_{c,\bar{y}}(x) = \sum_\nu \big((a_\nu \bar{y}) \odot (b_\nu x) + (b_\nu \bar{y}) \odot (a_\nu x)\big).$$

Its kernel is contained in the A-line $A\bar{y}$. Indeed, if $x \notin A\bar{y}$, project to the cross summand between the disjoint lines $A\bar{y}$ and Ax. After identifying both lines with A, the projection is the ordinary symmetrization of c, which is nonzero because symmetrization is injective on $\mathrm{Sym}^2(A)$ in odd characteristic. Thus the coefficient map has rank at least $\dim_{\mathbb{F}_q} D - d = t$.

The event $b \in \ker T_y^\top$ forces a rank-t affine system in independent fresh symbols. Sequentially condition on nonpivot symbols and apply the maximum-atom bound to the pivots, proving (29). Finally,

$$q^{-\operatorname{rank} T_y} = q^{-K} |\ker T_y^\top| = q^{-K}\left(1 + \sum_{b \neq 0} \mathbf{1}_{\{b \in \ker T_y^\top\}}\right).$$

Taking expectations gives $q^{-K} + (1 - q^{-K})\rho_{\mathrm{byte}}^t$. Sum over $y \neq 0$ to obtain (30). The excess above the unavoidable baseline $(q^h - 1)q^{-K}$ is nonnegative, so Markov's inequality gives (31). The two displayed choices of D have dimensions $d(v-1)$ and dv, respectively. $\square$

We henceforth use the nearly ideal threshold

$$\boxed{\eta_{128}(h, K) = (q^h - 1)q^{-K} + 2^{-128}.} \tag{32}$$

This is not a heuristic claim of perfect rank. It is a theorem-backed allowance of only 2^{-128} above the exact full-rank baseline.

Corollary 6.3 (Numerical spectrum tails)**.** *For the cost-driving slices,* (31) *with* $\varepsilon = 2^{-128}$ *gives the following base-two failure exponents. The entry in each cell is the weaker official shape when two shapes share a level.*

Route	*Level I*	*Level III*	*Level V*
$\ell = 4$ *fast channel core*	-138.12	-237.86	-371.14
$\ell = 4$ *direct complete square*	-95.64	-178.39	-294.67
$\ell = 2$ *one-eigenblock core*	-130.17	-263.46	-396.74
$\ell = 2$ *complete two-block core*	-62.21	-161.50	-260.80

Thus even the weakest headline event fails with probability below 2^{-62}*. A fixed key can instead publish its exact projective rank histogram, which certifies* (32) *without the random-XOF idealization.*

6.2 Core-complete recovery and the adaptive theorem

After scalar extension, partition the residual equations into multidegree families $\mathcal{F}_1, \ldots, \mathcal{F}_r$. A count vector $\mathbf{k} = (k_1, \ldots, k_r)$ chooses k_j linear combinations from family j, with $\sum_j k_j = h$. It is *core-complete* if every point at which the full restricted Jacobian has rank h has a nonzero h-row minor with a count vector in the chosen family.

Lemma 6.4 (Random projection within equation families)**.** *Let $J_j \in E^{m_j \times h}$ contain the Jacobian rows in family j. Suppose some choice of k_j rows from each family stacks to a nonzero $h \times h$ minor. If $P_j \in E^{k_j \times m_j}$ is uniform, then*

$$\Pr\left[\det\begin{pmatrix} P_1 J_1 \\ \vdots \\ P_r J_r \end{pmatrix} = 0\right] \leq \frac{h}{|E|}.$$

Proof. The determinant is a polynomial of total degree at most h in the entries of the P_j. It is not identically zero because coordinate-selector matrices recover the assumed minor. Apply Schwartz–Zippel. $\square$

Lemma 6.5 (Conservative well-separating-form success)**.** *For a square multihomogeneous core with Bézout number B, represented over a finite field E large enough for the finite-cardinality start system, the well-separating-form construction has a set of at most B^2 nonzero forbidden coefficient vectors. A uniform linear form therefore satisfies every separation condition with probability at least*

$$\sigma_E(B) = 1 - \frac{B^2}{|E|}.$$

Conditional on this event, the finite-cardinality symbolic-homotopy algorithm returns a zero-dimensional parametrization containing every nonsingular core root, with one-sided error.

Proof. The finite-cardinality homotopy construction reduces failure of its well-separating element to vanishing on a collection of at most B^2 nonzero vectors over E. For each fixed nonzero vector, a uniform coefficient vector annihilates it with probability at most $|E|^{-1}$. The union bound proves the displayed probability. We deliberately retain this theorem-level B^2 bound: ordinary pairwise separation alone would involve fewer hyperplanes, but it does not by itself verify every condition required by the cited homotopy parametrization. $\square$

Theorem 6.6 (Core-complete Las Vegas recovery)**.** *Assume (32) and let $\mathcal{P}$ be a finite core-complete family. Run every count vector in $\mathcal{P}$ with fresh within-family projections and a valid finite-cardinality homotopy field. Filter every parametrization through all residual equations, base-field descent, the signature rejection rule, affine reconstruction, and the original verifier. If $W_{\mathbf{k}}$ is the unit-leading invocation cost before root, projection, and separator success factors, then*

$$\mathbb{E}W \leq \frac{\sum_{\mathbf{k}\in\mathcal{P}} W_{\mathbf{k}}}{p_{\mathrm{ns}}\ \min_{\mathbf{k}}(1 - h/|E_{\mathbf{k}}|)\ \min_{\mathbf{k}}\left(1 - \frac{B_{\mathbf{k}}^2}{|E_{\mathbf{k}}|}\right)}. \tag{33}$$

The algorithm is Las Vegas and assumes no selected Macaulay degree, exact corank, unique root, or predetermined nonzero Jacobian core. Its actual work is κ_{hom} times the displayed ledger.

Proof. By Theorem 6.1, a trial contains an accepted full-Jacobian root with probability at least p_{ns}. At such a root, core completeness supplies a count vector with a nonzero minor. The family-projection lemma preserves it with the displayed conditional probability, and the separating form succeeds with the second displayed conditional probability. The corresponding homotopy invocation then contains the root in its output. Running every count vector guarantees that this invocation is present. Summing invocation costs and inverting the common success lower bound proves (33). Complete public filtering prevents an invalid output. $\square$

A *fast-core preflight* is an exact public test proving that the highest homogeneous part of the selected core's Jacobian determinant is not the zero polynomial. In the applications below, evaluating that highest part at one fixed public point suffices.

Theorem 6.7 (Adaptive certified-core fallback)**.** *Fix a public construction for which a common structural and spectrum event $\mathcal{S}$ makes a core-complete fallback valid. Suppose:*

1. *on passing an exact public fast-core preflight, the fast route has expected unit-leading work at most W_f;*
2. *on failing it, the core-complete fallback has expected work at most W_F;*
3. *under the random-XOF key distribution, the preflight fails with probability at most δ, while $\Pr[\mathcal{S}^c] \le \varepsilon$.*

For every fixed key in $\mathcal{S}$, the algorithm "test, then choose fast or fallback" is selected-degree-free and fixed-core-free. Under the random-XOF distribution, its conditional expected work obeys

$$\mathbb{E}[W \mid \mathcal{S}] \le W_f + \min\left\{1, \frac{\delta}{1-\varepsilon}\right\}(W_F - W_f)_+. \tag{34}$$

If an independent exact structural preflight succeeds with probability p_{str}, the random-key-normalized ledger divides the right side by $p_{\mathrm{str}}(1-\varepsilon)$. Without independence one may instead use any proved lower bound on $\Pr[\mathcal{S}]$.

Proof. On a fixed key, passing the preflight is a certificate, not a probabilistic premise. If it passes, the fast theorem applies; if it fails, core completeness applies. Thus no preferred core is assumed. For the expected work, let F be preflight failure. Then

$$\Pr[F \mid \mathcal{S}] \le \frac{\Pr[F]}{\Pr[\mathcal{S}]} \le \min\{1, \delta/(1-\varepsilon)\}.$$

Conditioning on the two branches gives $(1-q)W_f + qW_F = W_f + q(W_F - W_f)$ when $W_F \ge W_f$; if $W_F < W_f$, always using the fallback only helps, yielding the positive-part form. The last statement is the ordinary normalization by the density of generated keys that reach the solver. ☐

7 Constructive arithmetic and the homotopy ledger

The primary ledger below charges a complete extension-field multiplication circuit, including its fixed linear transforms. Additions, inversions, factorization, control, memory, and all remaining hidden constants stay in κ_{hom}.

Theorem 7.1 (Constructive 150-gate scalar multiplication)**.** *The circuit model assigns unit cost to two-input AND, XOR, and XNOR gates, with free constants, wires, and fan-out. For canonical five-bit representatives, the following complete circuit sizes are constructive:*

	addition	*negation*	*subtraction*	*multiplication*
$\mathbb{F}_{19}$	42	16	58	150
$\mathbb{F}_{19^2}$	84	32	116	692
$\mathbb{F}_{19^4}$	168	64	232	2628.

The 150*-gate scalar netlist is correct on all* $19^2 = 361$ *canonical input pairs. The tower representations are*

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2+1), \qquad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2-(1+u)).$$

They are exactly isomorphic to the manuscript basis $\mathbb{F}_{19}[t]/(t^4 - t - 1)$ *via*

$$u = 1 + 2t + 15t^2 + 5t^3, \qquad v = 4 + 13t + 13t^2 + t^3,$$

and the basis $(1, u, v, uv)$ *has determinant* $16 \neq 0$ *modulo* 19.

Proof. For a canonical scalar $x = (x_0, \dots, x_4)$, the circuit first converts to balanced signed magnitude. On the valid inputs $0, \dots, 18$, set

$$a = x_1 x_3, \quad b = x_2 x_3, \quad s = a \oplus b \oplus a x_2 \oplus x_4,$$

$$(m_0, m_1, m_2, m_3) = (x_0 \oplus s, x_1 \oplus s, x_2, x_3 \oplus b).$$

A direct truth-table check gives $s = 0, m = x$ for $x \leq 9$ and $s = 1, m = 19 - x$ for $10 \leq x \leq 18$. The two decoders use 18 gates. An exact four-by-four magnitude product together with the output sign reaches gate 80. Since the magnitude product is at most 81, write it as $\ell + 32h$ with $h \in \{0, 1, 2\}$, replace $32h$ by $13h$ mod 19, fold once, and reduce the reachable five-bit values; this reaches gate 130. A 20-gate signed recoding maps r to either r or $-r$ mod 19. The released netlist contains all 150 gates and an independent evaluator checks every canonical pair and canonical output.

For $u^2 = -1$, Karatsuba uses three scalar products, two input sums, and shared output negations, giving

$$5 \cdot 42 + 2 \cdot 16 + 3 \cdot 150 = 692.$$

At the second tower level, three $\mathbb{F}_{19^2}$ products cost $3 \cdot 692$. The two input sums cost $2 \cdot 84$, multiplication by $1 + u$ costs one scalar addition and subtraction, the low output costs one $\mathbb{F}_{19^2}$ addition, and the high output forms $p_0 + p_1$ once and subtracts it from p_2. Hence

$$2 \cdot 84 + 3 \cdot 692 + (42 + 58) + 84 + (84 + 116) = 2628.$$

Direct polynomial arithmetic verifies $u^2 = -1$, $v^2 = 1 + u$, the displayed coordinates, and the basis determinant. $\square$

For a core over $A \in \{\mathbb{F}_{19^2}, \mathbb{F}_{19^4}\}$ with Bézout number B and start-node requirement D_{start}, let g_A be 692 or 2628 and define

$$\Gamma_A(B, D_{\text{start}}) = \min_{\substack{e \geq 1:\ |A|^e \geq D_{\text{start}},\ |A|^e > B^2 \\ 2e-1 \leq |A|}} \frac{(2e-1) g_A}{1 - B^2/|A|^e}. \tag{35}$$

Evaluation/interpolation at $2e - 1$ base-field points gives the number of charged pointwise multiplications in the numerator; the fixed transforms themselves remain in κ_{hom}. Theorem 6.5 gives the denominator. The numerical artifact exhausts all admissible e.

8 The direct complete-square theorem

The following specialization is the main conceptual simplification. It uses all residual equations on a square slice, so no Jacobian-core choice or family-count enumeration remains.

Theorem 8.1 (Direct complete-square recovery)**.** *Retain the setting of Theorem 6.1 and suppose* $h = K$. *Restrict the complete residual map* g *to the public affine* K*-dimensional slice and solve all*

K equations together. On the event $\bar{\eta}_L^U \le \eta$, a trial contains an accepted root at which the Jacobian of this very system is nonsingular with probability at least

$$p_\square = \frac{a^2}{a+\eta}, \qquad a = \alpha - \frac{\eta}{q-1}.$$

If every equation has ordinary degree at most two, then

$$B_\square = 2^K, \qquad B_\square^+ = 2^K\left(1 + \frac{K}{2}\right).$$

Let

$$D_K = \binom{K+1}{2}, \qquad L_\square = D_K + 2K(D_K + K + 1), \qquad H_\square = L_\square + 2K + K^2.$$

Over any validated finite-cardinality homotopy field, the unit-leading work is

$$W_\square = \frac{\Gamma_A(2^K, 2K)}{p_\square} 2^{2K}\left(1 + \frac{K}{2}\right) H_\square K. \tag{36}$$

The algorithm is Las Vegas after complete public filtering and assumes no selected Macaulay degree, exact corank, unique root, preferred core, or orbit sweep.

Proof. With $h = K$, Theorem 6.1 applies directly to the Jacobian of the complete residual system; no projection to a subcore is made. The ordinary total-degree Bézout number of K quadratics in K variables is 2^K. In the companion homotopy coefficient, selecting the auxiliary term from no factor contributes 2^K, while selecting it from one of the K degree-two factors contributes 2^{K-1}; hence $B_\square^+ = 2^K + K2^{K-1}$. The displayed straight-line program first forms all D_K quadratic monomials and then evaluates the K affine quadratics, charging a coefficient multiplication and addition for every possible term. The homotopy theorem and Theorem 6.5 give (36). Every point returned by the parametrization is checked against the full residual system, affine reconstruction, rejection rule, and original verifier. □

Corollary 8.2 (Direct square attacks for all six $\ell = 4$ rows)**.** *For every official $\ell = 4$ shape, any public key passing the exact quotient and affine-rank preflights leaves a fresh ordinary K-dimensional slice in the public-vinegar region. Applying Theorem 8.1, embedding the $\mathbb{F}_{19}$ coefficients into $\mathbb{F}_{19^2}$, and charging the complete* 692*-gate multiplication circuit gives normalized exponents*

$$\boxed{142.60696, \qquad 185.45088, \qquad 227.56041}$$

on the worse shape at Levels I, III, and V. The corresponding nominal headroom below 143, 207, 272 *is*

$$\boxed{0.39304, \qquad 21.54912, \qquad 44.43959}$$

bits. Thus the six $\ell = 4$ rows are below their numerical references even without a fast-core preflight or the $55 \to 16$ orbit reduction.

Proof. The manuscript's fresh-slice inequality gives $\dim(V' \cap \ker C_{R,v}) \ge K$ for all six rows, so a cross-data-selected ordinary K-slice exists. The full-domain spectrum theorem applies with $t = 4(v-2)$ and the threshold $\eta_{128}(K, K)$. Substitute $K = 50, 70, 90$ in (36). Exact enumeration of the separator extension degree selects $e = 13, 17, 22$ over $\mathbb{F}_{19^2}$. The machine-readable ledger conditions on the exact structural preflight, normalizes only the certified spectrum event, and charges the conservative square-row acceptance lower bound. □

9 The zero-offset $\ell = 2$ theorem

For the three official $\ell = r = 2$ rows put

$$A = \mathbb{F}_{19^2}, \qquad m = m_1 = o, \qquad M = 4m, \qquad K = 3m.$$

Use the zero-offset column relation

$$u_0 = u, \qquad u_1 = 0. \tag{37}$$

Proposition 9.1 (Exact zero-offset target split)**.** *Let Q_R be the quotient-output matrix for (37). The route applies to a fixed key only after the exact public preflight* $\operatorname{rank} Q_R = K$*. This revision does not replace that fixed-key test by an unproved random-key density assertion for the three $\ell = 2$ rows. After a passing preflight, choose a left inverse G_R and a full-row-rank matrix W_R spanning the left kernel of Q_R. The complete substituted verifier is equivalent to*

$$W_R t = 0, \qquad z_{\mathrm{sym}}(u) = G_R t. \tag{38}$$

For a uniform official target, consistency has probability $19^{-(M-K)} = 19^{-m}$, and conditioned on it the residual target is uniform in $\mathbb{F}_{19}^K$. Chosen-message and salt inputs can therefore be filtered before solving. If one deliberately charges 2^{32} AXN operations per retained exactly uniform target, the additive target-generation exponents are

$$16 \log_2 19 + 32 = 99.96684, \quad 24 \log_2 19 + 32 = 133.95026, \quad 32 \log_2 19 + 32 = 167.93368.$$

Proof. Zero offsets make every affine and constant contribution vanish, so the substituted verifier is $Q_R z_{\mathrm{sym}}(u) = t$. Applying the invertible output change with block rows G_R, W_R gives (38). An invertible linear transformation preserves uniformity and separates the quotient and consistency coordinates. Target generation occurs before the nonlinear solve, so its work adds to rather than multiplies the solve ledger. □

In one block, (37) has matrix $\left(\begin{smallmatrix} x & 0 \\ y & 0 \end{smallmatrix}\right)$ and is symmetric exactly when $y = 0$. The $n = v + o$ block conditions use disjoint coordinates. Thus the exact density accepted by the Version 2.3 rule is

$$\alpha_n = 19^{-n} \sum_{j=0}^{\lfloor n/4 \rfloor} \binom{n}{j} 18^{n-j}. \tag{39}$$

Choose all solver slices inside the full public-vinegar summand A^v, independently of its fresh internal symmetric coefficients. By Theorem 6.2, the fresh exponent is $t = 2(v-1)$.

9.1 The complete two-block core

Take an A-linear slice of dimension $s = 3m/2$. Its base-field dimension is $h = 2s = K$, and the complete quotient system has m equations in each bidegree family

$$(2, 0), \qquad (1, 1), \qquad (0, 2).$$

It is therefore square and core-complete without selecting or discarding any residual equation.

Theorem 9.2 (Complete two-block $\ell = 2$ forger)**.** *On the event $\bar{\eta}_L^U \le \eta_{128}(K,K)$, a consistent target–coset trial contains an accepted full-Jacobian root with probability at least*

$$p^{(2)}_{\mathrm{comp}} = \frac{a_2^2}{a_2 + \eta_{128}(K,K)}, \qquad a_2 = \alpha_n - \frac{\eta_{128}(K,K)}{18}. \tag{40}$$

The exact multihomogeneous quantities are

$$B^{(2)}_m = 2^{2m}\binom{m}{m/2}, \tag{41}$$

$$(B^{(2)}_m)^+ = B^{(2)}_m\left(1 + 2m + \frac{m^2}{m+2}\right). \tag{42}$$

Finite-cardinality homotopy over $\mathbb{F}_{19^2}$, Frobenius descent, and complete public filtering form a selected-degree-free, fixed-core-free Las Vegas forger. The normalized primary exponents, including the additive target filter and the 2^{-128} spectrum normalization, are

$$132.97067, \qquad 183.79463, \qquad 233.84399$$

for Levels I, III, and V.

Proof. Apply Theorem 6.1 with $h = K$ and the exact accepted density (39). Over the splitting field, each base form supplies one equation in each of the two diagonal families and the cross family. The Bézout number is

$$[t_0^s t_1^s](2t_0)^m (t_0 + t_1)^m (2t_1)^m.$$

The cross factor must contribute $m/2$ powers to each block, proving (41). In the companion homotopy coefficient, the term with no auxiliary variable contributes B. Deleting a diagonal factor contributes $2^{2m-1}(\binom{m}{m/2} + \binom{m}{m/2+1})$, while deleting a cross factor contributes B. Summing and using $\binom{m}{m/2+1}/\binom{m}{m/2} = m/(m+2)$ proves (42). The exact separator factor is (35); every later operation is an exact filter. $\square$

Put

$$\begin{aligned} D_s &= \binom{s+1}{2}, & C_s &= (s+1)^2, \\ L_{\mathrm{comp}} &= 2D_s + s^2 + 4m(D_s + s + 1) + 2mC_s, \\ H_{\mathrm{comp}} &= L_{\mathrm{comp}} + 6m + K^2. \end{aligned} \tag{43}$$

The unit-leading work on a good spectrum event is

$$W^{(2)}_{\mathrm{comp}} = \frac{\Gamma_{\mathbb{F}_{19^2}}(B^{(2)}_m, 2s)}{p^{(2)}_{\mathrm{comp}}} B^{(2)}_m (B^{(2)}_m)^+ H_{\mathrm{comp}} K. \tag{44}$$

Divide by the exact spectrum-event density and add the target-generation ledger.

9.2 The faster Level-I core and its exact fallback

For $s \le m$, select s diagonal equations and solve one eigenblock of A^s. A descended full root reconstructs the other block by Frobenius. The core is a square system of s ordinary quadratics, with

$$B_s = 2^s, \qquad B_s^+ = 2^s(1 + s/2).$$

At a fixed nonzero public point, the fresh leading row of each selected public form is surjective onto $(A^s)^\vee$. Sequential exposure therefore gives

$$p^{(2)}_{\mathrm{Jac}}(s) \ge \prod_{j=1}^{s} (1 - \rho^{2j}_{\mathrm{byte}}) > 0.9970003329. \tag{45}$$

A nonzero leading determinant has degree at most s over A, so its singular point density is at most $s/19^2$. With $\eta_s = \eta_{128}(2s, 3m)$, a target–coset trial has an accepted full root nonsingular for this core with probability at least

$$p_s = 19^{2s-3m} \max \left\{ a_s - \eta_s, \frac{a_s^2}{a_s + \eta_s} \right\}. \tag{46}$$

where $a_s = \alpha_n - s/19^2$. These are respectively the singleton and second-moment lower bounds after subtracting the core-singular locus.

At Level I take $s = m = 16$. Run the exact leading-determinant preflight. If it passes, solve the one-eigenblock core; if it fails, solve the complete two-block system of Theorem 9.2. By Theorem 6.7, this is fixed-core-free by construction and has normalized exponent

$$\boxed{132.18807}. \tag{47}$$

the diagonal-only and complete-only exponents in the strengthened ledger are 132.18928 and 132.97067. The adaptive value is slightly smaller than the first rounded figure because the common spectrum normalization is charged once and the exact branch mixture is performed before taking logarithms. At Levels III and V, the complete core is already faster.

10 An explicit Boolean-ledger Level-I route via Just Guess

The homotopy ledger deliberately leaves κ_{hom} explicit. This section asks whether the numerically weakest row, $(v, o, \ell, r) = (48, 16, 2, 2)$, admits a more concrete estimate. We combine the exact zero-offset target split of Theorem 9.1 with the underdetermined-MQ transformation of May, Ostuzzi, and Ressler [20]. Their analysis reduces a random underdetermined MQ system to guessed variables, a short tree of univariate quadratics, a linear solve, and check equations. It explicitly uses a random-system regularity model for the linearization step. We retain that premise rather than turning it into an unconditional statement.

Write the quotient map in its two Frobenius channels as

$$D : A^{64} \longrightarrow A^{16}, \qquad H : A^{64} \longrightarrow \mathbb{F}_{19}^{16}, \qquad A = \mathbb{F}_{19^2}.$$

Here D is the diagonal channel and H is the Hermitian cross channel. The retained consistent target is (t_D, t_H). We additionally reject the event $t_D = 0$, of probability 19^{-32} under the exact uniform target sampler. Conditioned on consistency and $t_D \ne 0$, the pair (t_D, t_H) remains exactly uniform over the retained target set.

The two channel families are distinct linear images of the same native byte-reduced public coefficients. They are therefore *not independent*. The argument below uses an exact conditional atom bound for the official $\ell = 2$ channel transform instead.

10.1 The streamed diagonal solver

Apply Just Guess to the $m = 16$ equations $D(x) = t_D$ in $n = 64$ variables over A, with

$$k = 5, \qquad p = 6, \qquad m - k - p = 5.$$

The published structural inequalities hold:

$$2k + p = m, \qquad 2(n - m) = 96 \geq 6(2m - 2k - p - 1) = 90,$$

$$n - 1 = 63 \geq (m - k - p - 1)(m - k + 2) = 52.$$

The public transformation gives coordinates $(u, z) \in A^{16} \times A^{48}$. Fixing z changes only linear and constant terms in the transformed square system. We stream an A^8 family of z assignments. For each assignment, the solver enumerates all $|A|^5$ guesses, follows every returned quadratic branch, solves the remaining 5×5 linear system, and checks the five withheld equations. Surviving points are tested against $H(x) = t_H$, the format rule, affine reconstruction, and the unmodified verifier.

Let $\mathcal{R}$ denote the event that this computation supplies at least $19^{16}/4$ distinct sign-canonical diagonal roots. This is a concrete, transcript-checkable form of the published Just Guess random-system regularity premise. We write κ_{JG} for the expected multiplier needed to obtain $\mathcal{R}$. It includes transformation and linearization retries, candidate-abundance failures, the negligible format and $t_D = 0$ losses, and any candidate-processing cost beyond the explicit reserve below. Unlike κ_{hom}, it does not hide a generic soft-O algebraic solver, but it remains an unmeasured premise.

10.2 Conditional cross-channel filtering

For $x \in A^{64}$ put $\Psi(x) = x x^{19\top}$. If $\Psi(x) = \Psi(y)$ for nonzero x, y, then $y = \lambda x$ with $\lambda\lambda^{19} = 1$. If also $D(x) = D(y) = t_D \neq 0$, then $\lambda^2 = 1$, so $y = \pm x$. Retaining one representative of each sign pair therefore makes the cross tensors distinct.

Lemma 10.1 (Exact conditional channel atom)**.** *Use the native-coordinate random-XOF model, in which one field symbol has raw integer weight* 14 *for residues* $0, \ldots, 8$ *and* 13 *for residues* $9, \ldots, 18$*. Condition on the complete diagonal-channel coefficient family* D*. For every native coefficient block and every nonzero* $\mathbb{F}_{19}$*-linear functional of its remaining Hermitian coefficient, every conditional atom has probability at most*

$$\rho_{\mathrm{cond}} = \frac{49}{829} < 0.059108. \tag{48}$$

Conditioned on D*, different native blocks and different base forms remain independent.*

Proof. Diagonalize the official matrix $S = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 15 \end{smallmatrix}\right)$ over $A = \mathbb{F}_{19}[u]/(u^2 + 1)$. Its eigenvalues are $8 \pm 2u$, with eigenvectors $(1, 13 \pm u)$. For a native symmetric diagonal block $\left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$, the diagonal coefficient and Hermitian coefficient are

$$D = (a + 7b + 16c) + (2b + 7c)u, \qquad H = a + 7b + 18c.$$

For an off-diagonal native block $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, they are

$$D = (a + 13b + 13c + 16d) + (b + c + 7d)u,$$

$$H = (a + 13b + 13c + 18d) + (18b + c)u.$$

These maps act separately on disjoint native blocks. Exact enumeration of all 19^3 diagonal tuples with their product weights gives maximum conditional atom $49/829$. Exact enumeration of all

19^4 off-diagonal tuples and all 20 projective nonzero linear functionals on the A-valued Hermitian coefficient gives the smaller maximum 169246/2971565. The artifact reproduces both rational values directly from the displayed formulae. Conditioning separately on the D-image of each independent native block preserves blockwise and base-form independence. □

Lemma 10.2 (Conditional cross-channel second moment)**.** *Let C be any set of $T = c\,19^{16}$ sign-canonical solutions of $D(x) = t_D$, where $t_D \neq 0$ and $0 < c \leq 1$. Condition on the entire diagonal channel and on any solver transcript determined by it. For a uniform independent target t_H,*

$$\Pr[\exists x \in C : H(x) = t_H \mid D] \geq \frac{c}{1 + c(19\rho_{\mathrm{cond}})^{16}}. \tag{49}$$

For $c = 1/4$, this lower bound is $0.0961344407\ldots$.

Proof. Let $N = |\{x \in C : H(x) = t_H\}|$. Uniformity of t_H gives $\mathbb{E}N = c$. For distinct $x, y \in C$, the Hermitian tensors differ. For each of the sixteen independent base forms, the equality $H_i(x) = H_i(y)$ is a nonzero linear equation in the Hermitian block coefficients. Condition on all blocks except one on which this functional is nonzero. By Theorem 10.1, its conditional probability is at most ρ_{cond}. Independence across the sixteen base forms gives

$$\Pr[H(x) = H(y) \mid D] \leq \rho_{\mathrm{cond}}^{16}.$$

Consequently

$$\mathbb{E}N(N-1) \leq T(T-1)19^{-16}\rho_{\mathrm{cond}}^{16} < c^2(19\rho_{\mathrm{cond}})^{16}.$$

Cauchy–Schwarz gives $\Pr[N > 0] \geq (\mathbb{E}N)^2/\mathbb{E}[N^2]$, proving the claim. □

10.3 Finite Boolean-gate instantiation

We use the verified 692-gate multiplier and 84-gate adder over A. A monic quadratic over A is solved by a fixed Tonelli–Shanks schedule. The artifact checks all 361 possible discriminants and obtains

$$C_{\mathrm{root}} = 16{,}886 \quad \mathsf{AXN}\text{ gates.} \tag{50}$$

Instantiating the published expected Just Guess operation formulas at $(n, m, k, p) = (64, 16, 5, 6)$ gives, per guess,

2684.089 extension multiplications, 2621.087 extension additions, 6 quadratic-root calls.

Rounding upward and adding one million gates for comparisons, address logic, and branch control gives

$$C_{\mathrm{guess}} = 3{,}179{,}584 \quad \mathsf{AXN}\text{ gates.} \tag{51}$$

We declare a reserve of 2^{40} gates per z assignment for candidate filtering and verifier calls. This is a ledger allowance, not a theorem that it dominates every possible candidate population; any excess is included in κ_{JG}. Target sampling, the public transformation, and all rank preflights are polynomial-sized additive terms.

Proposition 10.3 (Finite-gate Level-I sensitivity)**.** *Under the published Just Guess expected-operation model, the Boolean ledger for one A^8-streaming trial is*

$$W_{\mathrm{JG}}^{\mathrm{trial}} = |A|^{8+5}C_{\mathrm{guess}} + |A|^8 2^{40} < 2^{132.047}. \tag{52}$$

On event $\mathcal{R}$, Theorem 10.2 with $c = 1/4$ gives an expected cross-channel restart multiplier below 10.403*. Consequently the success-adjusted Boolean-gate ledger is*

$$\log_2 W_{\mathrm{JG}} < 135.426 + \log_2 \kappa_{\mathrm{JG}}. \tag{53}$$

It is below the Level-I reference 143 *whenever*

$$\log_2 \kappa_{\mathrm{JG}} < 7.575. \tag{54}$$

*For orientation only, a factor-*16 *regularity reserve gives exponent* 139.426 *and* 3.574 *bits of remaining margin. A separate capped-tree schedule follows at most* 63 *quadratic nodes and* 64 *leaves per guess and aborts any branch whose residual linear system is not unique. Its trial exponent is* 137.714*; after the same cross-channel restart it is $141.093 + \log_2 \kappa_{\mathrm{JG}}^{\mathrm{cap}}$. Thus this strictly bounded-work sensitivity remains below* 143 *when $\log_2 \kappa_{\mathrm{JG}}^{\mathrm{cap}} < 1.908$. It is reported as a sensitivity check, not as a proof of the capped candidate event.*

Proof. There are $|A|^8$ streamed assignments and $|A|^5$ guesses per assignment. Substituting the verified field circuits and quadratic-root schedule into the published expected-operation formulas gives (52); the artifact recomputes the integer ledger before taking logarithms. On $\mathcal{R}$, take any $19^{16}/4$ sign-canonical candidates. The reciprocal of (49) is $10.4020993\ldots$, adding 3.378803 bits. The remaining regularity, transformation, candidate-collection, and excess filtering multipliers are, by definition, included in κ_{JG}. Every candidate is checked by the original verifier, so all failure modes are one-sided. $\square$

Interpretation. This proposition materially narrows the referee's cost-model objection but does not erase it by notation. It replaces the homotopy soft-O multiplier on the weakest row by (i) an explicit Boolean ledger for the published expected-operation schedule and (ii) the specific, publicly auditable event that the Just Guess run yields enough diagonal roots. It does not provide a complete circuit for every control path or prove candidate abundance. A production-size fixed-key transcript would measure κ_{JG} directly. The two $\ell = 4$ Level-I rows still retain κ_{hom} in their homotopy ledgers.

11 Adaptive $\ell = 4$ recovery

Let $A = \mathbb{F}_{19^4}$ and let $L \cong A^s$ be the public slice. Under the Frobenius-distance normal form, each public base form supplies an A-valued degree-two channel and an A-valued degree-20 channel. Write their highest homogeneous parts as

$$F_{i,0}^{\mathrm{top}}(x) = C_{i,0}(x, x), \qquad F_{i,1}^{\mathrm{top}}(x) = C_{i,1}(x, x^{19}).$$

Lemma 11.1 (Fresh channel decomposition and row surjectivity)**.** *Restriction of a fresh symmetric $\mathbb{F}_{19}$-bilinear form to the injected public-vinegar copy of L, followed by the Frobenius-distance transform, is surjective onto*

$$\mathrm{Sym}_A^2(L^\vee) \oplus \mathrm{Bil}_A(L, L^{(19)}; A).$$

For every fixed $0 \neq x_\star \in L$, the map from one public form's fresh coefficients to the pair of leading Jacobian rows

$$(DF_{i,0}^{\mathrm{top}}(x_\star), DF_{i,1}^{\mathrm{top}}(x_\star)) \in L^\vee \oplus L^\vee$$

is surjective over A.

Proof. After extending scalars, the four diagonal blocks of a symmetric form form one Frobenius orbit and descend to an arbitrary symmetric A-bilinear form; the four adjacent blocks form a second orbit and descend to an arbitrary bilinear form between L and its Frobenius twist. The orbit coefficient sets are disjoint, so the two choices are independent. Formal differentiation gives

$$DF_{i,0}^{\mathrm{top}}(x_\star)[h] = 2C_{i,0}(x_\star, h), \qquad DF_{i,1}^{\mathrm{top}}(x_\star)[h] = C_{i,1}(h, x_\star^{19}).$$

Evaluation of arbitrary bilinear forms is surjective. For the symmetric case, given $\lambda \in L^\vee$, choose $\phi(x_\star) = 1$ and use

$$\frac{1}{2}(\phi \otimes \lambda + \lambda \otimes \phi - \lambda(x_\star)\phi \otimes \phi).$$

$\square$

Choose a whole coordinates of the degree-two channel and b whole coordinates of the degree-20 channel, with $a + b = s$. The determinant preflight is exact on the public key. For its random-XOF density we use the following conservative bound, which does not assume that an invertible linear change preserves independence under the slightly biased modulo-19 distribution.

Lemma 11.2 (Projective Jacobian-preflight density)**.** *At the fixed public preflight point, let $J \in A^{s\times s}$ be the selected leading-Jacobian matrix, where $A = \mathbb{F}_{19^4}$. Then*

$$\Pr[\det J = 0] \le \frac{|A|^s - 1}{|A| - 1}\rho_{\mathrm{byte}}^{4s}. \tag{55}$$

For the cost-minimizing values $s = 10, 14, 18$, the failure bounds are at most $3.55350 \cdot 10^{-5}$, $6.56029 \cdot 10^{-5}$, and $1.21113 \cdot 10^{-4}$, respectively.

Proof. If J is singular, some projective direction $[y] \in \mathbb{P}^{s-1}(A)$ lies in its right kernel. Fix such a direction. The joint channel-surjectivity statement above implies that forcing every selected leading row to annihilate y imposes s independent equations over A, hence $4s$ independent affine equations over $\mathbb{F}_{19}$, on fresh public-expansion symbols. The maximum-atom bound gives probability at most $\rho_{\mathrm{byte}}^{4s}$. There are $(|A|^s - 1)/(|A| - 1)$ projective directions. Union-bound over them. This bound is weaker than a sequential full-rank product would be under a uniform field distribution, but it is valid for the actual biased product distribution used in the random-XOF model. $\square$

Conditional on passing the preflight, put

$$a_{a,b} = \alpha - \frac{a + 19b}{19^4}.$$

The sharpened core-root probability is

$$p_{a,b,s} = 19^{4s-K}\frac{a_{a,b}^2}{a_{a,b} + \eta_{128}(4s, K)}. \tag{56}$$

Its exact one-block Bézout data are

$$B = 2^a 20^b, \qquad B^+ = B(1 + a/2 + b/20).$$

With

$$L_{a,b,s} = \mathbf{1}_{a>0}\left[\binom{s+1}{2} + 2a\left(\binom{s+1}{2} + s + 1\right)\right]$$

$$+\mathbf{1}_{b>0}\left[6s+s^2+2b(s^2+2s+1)\right],$$

the unit-leading good-key work is

$$\frac{\Gamma_{\mathbb{F}_{19^4}}(B,2a+20b)}{p_{a,b,s}}BB^+(L_{a,b,s}+2a+20b+s^2)s. \tag{57}$$

Exhaustive enumeration of all $35, 63, 99$ admissible profiles gives

$$(s;a,b)=(10;5,5),\quad (14;7,7),\quad (18;9,9)$$

at Levels I, III, V. Conditional on the exact structural preflight, the worse-shape fast-core ledger exponents are 126.260, 167.038, and 207.273.

For the simplest fixed-core-free algorithm, a failed preflight invokes the direct ordinary square of Theorem 8.2. Applying Theorem 6.7 gives:

Level	parameters	fast profile	fast conditional	adaptive mixture	headroom
I	$(28,5,4,4)$	$(10;5,5)$	126.260	128.246	14.754
I	$(28,4,4,5)$	$(10;5,5)$	126.260	128.246	14.754
III	$(40,7,4,4)$	$(14;7,7)$	167.038	171.617	35.383
III	$(38,5,4,5)$	$(14;7,7)$	167.038	171.616	35.384
V	$(50,9,4,4)$	$(18;9,9)$	207.273	214.558	57.442
V	$(52,6,4,6)$	$(18;9,9)$	207.273	214.557	57.443

This table needs no family-count sweep. Earlier drafts also reported a sixteen-orbit optimization. Because the released package did not contain its per-orbit completeness and cost records, that numerical branch is omitted from this revision; it is unnecessary for every headline conclusion.

The same profile search can be optimized for output size rather than arithmetic. We omit those secondary numerical rows here because the revised standalone generator records only the direct and adaptive headline routes.

12 All-nine consequences

Theorem 12.1 (One complete-square theorem for every published row)**.** *For every published $q=19$ Version 2.3 parameter shape, a public key passing the stated exact rank preflights and satisfying the certified spectrum event (or carrying the corresponding fixed-key certificate) has a selected-degree-free Las Vegas forgery route obtained by solving a complete square residual system. It uses no selected Macaulay degree, exact-corank prediction, unique-root assumption, predetermined Jacobian core, family-count pattern, or orbit enumeration. The unit-leading homotopy ledgers are:*

Level	*parameters*	*complete square*	*exponent*	*headroom*
I	$(28,5,4,4)$	*ordinary* 50×50 *square*	142.607	0.393
I	$(48,16,2,2)$	*complete two-block square*	132.971	10.029
I	$(28,4,4,5)$	*ordinary* 50×50 *square*	142.606	0.394
III	$(40,7,4,4)$	*ordinary* 70×70 *square*	185.451	21.549
III	$(72,24,2,2)$	*complete two-block square*	183.795	23.205
III	$(38,5,4,5)$	*ordinary* 70×70 *square*	185.450	21.550
V	$(50,9,4,4)$	*ordinary* 90×90 *square*	227.560	44.440
V	$(96,32,2,2)$	*complete two-block square*	233.844	38.156
V	$(52,6,4,6)$	*ordinary* 90×90 *square*	227.559	44.441

The worst rows at Levels I, III, and V have exponents

$$\boxed{142.60696, \qquad 185.45088, \qquad 233.84399}. \tag{58}$$

Thus even this deliberately uniform proof is nominally below the 143, 207, 272 *references at all three levels.*

Proof. The $\ell = 2$ rows are Theorem 9.2; the $\ell = 4$ rows are Theorem 8.2. Both routes use the same accepted-root theorem, exact separator ledger, and complete public filtering. The displayed worst values are the levelwise maxima. □

Theorem 12.2 (Optimized adaptive attack on every published row)**.** *Under the same structural and spectrum conditions, an exact public determinant preflight can replace the direct square by a smaller core without introducing a preferred-core premise: failure invokes the direct complete square. The resulting all-nine table is*

Level	*parameters*	*adaptive selected-degree-free route*	*exponent*	*headroom*
I	$(28, 5, 4, 4)$	*channel core, then ordinary square*	128.246	14.754
I	$(48, 16, 2, 2)$	*one eigenblock, then two-block square*	132.188	10.812
I	$(28, 4, 4, 5)$	*channel core, then ordinary square*	128.246	14.754
III	$(40, 7, 4, 4)$	*channel core, then ordinary square*	171.617	35.383
III	$(72, 24, 2, 2)$	*complete two-block square*	183.795	23.205
III	$(38, 5, 4, 5)$	*channel core, then ordinary square*	171.616	35.384
V	$(50, 9, 4, 4)$	*channel core, then ordinary square*	214.558	57.442
V	$(96, 32, 2, 2)$	*complete two-block square*	233.844	38.156
V	$(52, 6, 4, 6)$	*channel core, then ordinary square*	214.557	57.443

Taking the worst row at each level gives

$$\boxed{132.18807, \qquad 183.79463, \qquad 233.84399}. \tag{59}$$

The corresponding nominal headroom is

$$\boxed{10.81193, \qquad 23.20537, \qquad 38.15601} \text{ bits.}$$

Equivalently, the adaptive comparison is below the numerical references only when

$$\log_2 \kappa_{\mathrm{hom}} < 10.81193, \qquad 23.20537, \qquad 38.15601$$

at Levels I, III, and V. No particular value of κ_{hom} *is asserted.*

Proof. Use the exact branch rule of Theorem 6.7. For $\ell = 2$, the Level-I fast branch is the certified one-eigenblock core and the fallback is the complete two-block square; the complete square is already faster at Levels III and V. For $\ell = 4$, use Theorem 11.2 and fall back to Theorem 8.2. The machine-readable ledger mixes branch work before taking logarithms, charges the union of the relevant spectrum tails, and is conditional on the exact structural preflight; only the common spectrum-event density is normalized probabilistically. □

Lower-output refinement. Deliberately using the one-eigenblock $\ell = 2$ core whenever its certificate passes yields a smaller-output fixed-core-free frontier with worst exponents

$$\boxed{132.18807, \qquad 185.43454, \qquad 237.78324}.$$

Its fast branch is taken with probability above 0.997; the complete square remains an exact fallback.

Scope of the arithmetic unit. The primary tables charge complete validated multiplication circuits for each pointwise product over the base or tower fields $\mathbb{F}_{19}$, $\mathbb{F}_{19^2}$, and $\mathbb{F}_{19^4}$. Extension-field evaluation/interpolation and basis transforms, together with additions, inversions, factorization, control, filtering, memory traffic, and implementation constants, remain inside κ_{hom}. A narrower variable-product-only normalization is retained only in the artifact and is not used for any theorem headline.

13 Complementary streamed-XL and low-state frontiers

The complete-square homotopy theorem is the logical headline, but it does not supply a low-memory implementation. This section records a complementary branch whose matrices can be streamed with linear vector state. Its correctness is exact once a finite quotient certificate is produced; its numerical exponent retains a production selected-degree premise. The two branches therefore answer different questions rather than competing for one headline.

13.1 Four-block special XL

For an A-linear $\ell = 4$ slice of A-dimension s, scalar extension gives four blocks of s variables. Each unordered pair of blocks carries m_1 independent quadratic equations. After adding one homogenizer per block, the Hilbert series used by the published special-XL model is

$$H_{m_1,s}(\mathbf{t}) = \frac{\prod_{a=1}^{4}(1-t_a^2)^{m_1}\prod_{1\le a<b\le 4}(1-t_at_b)^{m_1}}{\prod_{a=1}^{4}(1-t_a)^{s+1}}. \tag{60}$$

For a multidegree $\mathbf{d} = (d_1, \dots, d_4)$, the monomial-column count is

$$M_s(\mathbf{d}) = \prod_{a=1}^{4}\binom{s+d_a}{d_a}. \tag{61}$$

A negative coefficient of (60) is the selected-degree trigger in the cited model [8]. It predicts that a planted instance has the finite quotient needed for extraction; the theorem below makes the operator and state accounting exact but does not prove that production instances reach such a quotient at that degree.

The exact row count is

$$\begin{aligned} R_s(\mathbf{d}) = {} & m_1\sum_{a=1}^{4}\binom{s+d_a-2}{d_a-2}\prod_{c\ne a}\binom{s+d_c}{d_c} \\ & + m_1\sum_{1\le a<b\le 4}\binom{s+d_a-1}{d_a-1}\binom{s+d_b-1}{d_b-1}\prod_{c\notin\{a,b\}}\binom{s+d_c}{d_c}, \end{aligned} \tag{62}$$

where a binomial coefficient with a negative lower argument is zero.

Proposition 13.1 (Streaming the special-XL operator)**.** *Let $A_{\mathbf{d}}$ be the Macaulay matrix at multidegree $\mathbf{d}$, with $M = M_s(\mathbf{d})$ columns and $R = R_s(\mathbf{d})$ rows. Every row and transpose-row product can be generated from the public quadratic coefficients using $O(s^2)$ scratch space.*

Over an extension $E/\mathbb{F}_{19^4}$, choose a random diagonal $\Delta \in E^{R\times R}$ and put

$$B = A_{\mathbf{d}}^{\intercal}\Delta A_{\mathbf{d}}.$$

Then

$$\Pr[\ker B \neq \ker A_{\mathbf{d}}] \leq M/|E|.$$

A product by B is evaluated by one pass over the streamed rows and uses $O(M)$ resident field-vector state. Scalar Wiedemann over $E = \mathbb{F}_{19^8}$ requires fewer than

$$16R(s+1)^2 M \tag{63}$$

signed $\mathbb{F}_{19^4}$ multiply–accumulates per trial. A conservative cubic preconditioner over $\mathbb{F}_{19^{12}}$ gives the Level-I bound $32R(s+1)^2M$.

Proof. The two sums in (62) count multipliers of the diagonal and off-diagonal equation families. A homogenized quadratic row has at most $(s+1)^2$ terms, so rows and transpose products are generated on demand.

Factor $A_{\mathbf{d}} = UV$ at rank r. If $U^{\mathsf{T}}\Delta U$ is nonsingular, then $\ker(A_{\mathbf{d}}^{\mathsf{T}}\Delta A_{\mathbf{d}}) = \ker V = \ker A_{\mathbf{d}}$. By Cauchy–Binet, $\det(U^{\mathsf{T}}\Delta U)$ is a nonzero multilinear polynomial of total degree r in the diagonal entries. Schwartz–Zippel gives failure at most $r/|E| \leq M/|E|$.

Streaming first computes the row inner product and then accumulates the scaled row. In the quadratic extension, the sparse dot product and accumulation use at most four base-extension products per row term and the diagonal multiplication uses three more. Fewer than $3M$ normal-operator products generate the scalar Wiedemann sequence and reconstruct one null vector. Rounding the resulting constant gives (63); the cubic-extension calculation is analogous. □

13.2 Certified extraction beyond exact corank one

A one-dimensional Macaulay kernel is convenient but is not required for soundness.

Theorem 13.2 (Recovery from a finite commuting border quotient)**.** *Fix a target and slice over a finite field Ω, and let $I \subseteq \Omega[x_1, \ldots, x_h]$ be the ideal of the residual equations. Let $\mathcal{B}$ be a finite divisor-closed monomial set containing 1. Suppose the Macaulay row space supplies, for every border monomial $\beta \in \partial\mathcal{B}$, a certified relation*

$$g_\beta = \beta - \sum_{b \in \mathcal{B}} c_{\beta,b} b \in I.$$

Construct the formal multiplication matrices $M_1, \ldots, M_h$ on $\Omega\mathcal{B}$. If they commute, then the border relations generate a zero-dimensional ideal $J \subseteq I$, $\mathcal{B}$ is a basis of $\Omega[x]/J$, and every root of I occurs among the roots enumerated from the multiplication tables. FGLM conversion costs $O(h|\mathcal{B}|^3)$ field operations once the tables are available. Complete residual and public verification removes every extra point.

Proof. Commutativity makes $x_i \mapsto M_i$ an algebra homomorphism. Divisor closure identifies each basis monomial with its coordinate vector, and every border relation acts as zero. The border-division theorem therefore gives unique normal forms in $\operatorname{span}_\Omega \mathcal{B}$, so the relations form a border basis of a zero-dimensional ideal J. Because every relation is a Macaulay row combination from I, we have $J \subseteq I$ and hence $V(I) \subseteq V(J)$. Standard FGLM conversion and finite-field factorization enumerate the geometric points of J [10, 11]. □

A quotient of dimension larger than one is therefore a correctness fallback, not a free extension of the scalar-Wiedemann exponent. Obtaining all border relations, block-nullspace work, and resident multiplication-table state must fit the displayed ledger or be charged separately.

13.3 Three-plus-one completion

The fourth eigenblock can be completed linearly after solving the first three. Let E_{ab} be the m_1 equations supported on blocks a, b.

Theorem 13.3 (Exact three-plus-one completion)**.** *Suppose a certified first-stage solver returns a finite set containing every root of the six families E_{ab} with $1 \le a \le b \le 3$. For each partial root $p = (x_1, x_2, x_3)$, substitute into E_{14}, E_{24}, E_{34} and obtain*

$$A_p x_4 = b_p, \qquad A_p \in \Omega^{3m_1 \times s}.$$

Discard inconsistent systems. If $\operatorname{rank} A_p = s$*, solve for the unique fourth block and retain it exactly when $E_{44} = 0$. If a consistent system has rank below s, declare the trial incomplete unless a separately priced procedure enumerates all completions. Whenever no incomplete case occurs, the retained set contains every root of the complete four-block system. Final descent and verification make the procedure Las Vegas.*

Proof. Any complete root restricts to a first-stage root. At that partial root its true fourth block satisfies the displayed linear system. On a trial that declares no incomplete case, every consistent system encountered has full column rank. Hence the true fourth block is the unique completion, and the remaining diagonal family checks it. Conversely every retained point satisfies all ten families. □

13.4 Status of the low-state branch

The streamed operator, row count, and correctness fallbacks above remain useful complementary machinery. Earlier drafts printed three numerical low-state profiles, but the released artifact did not regenerate their selected multidegrees, certificate probability, and state figures from a single checker. Those numbers are therefore omitted here. No selected- degree-free headline depends on them. A future release should restore a numerical frontier only together with the full profile generator and a production border-basis or solve transcript.

14 Implementation, evidence, and reproducibility

The companion artifact separates theorem checks from experimental evidence. The primary selected-degree-free conclusions no longer rely on sampled production ranks or small-profile Macaulay behavior. Their machine-checkable inputs are the exact public dimensions, the random-XOF entropy calculation, the multihomogeneous coefficient formulas, and the explicit arithmetic circuits.

Table 4: Evidence hierarchy used in this revision.

Check	What is computed	What it establishes
Exact public preflight	Quotient, affine, projection, skew, and determinant ranks for a concrete key	Validity of the exact reduction and the selected solver branch on that key.
Random-XOF theorem	Maximum-atom and fresh-coefficient rank calculations	Density of keys satisfying the aggregate spectrum and structural events in the idealized expansion.
Fixed-key certificate	Projective rank histogram and determinant transcript	Exact replacement for the random-XOF spectrum and fast-core density statements on one key.
Homotopy ledger	Exact Bézout numbers, companion coefficients, field thresholds, and straight-line-program counts	Unit-leading arithmetic theorem; not a wall-clock or low-memory implementation.
Circuit certificate	Exhaustive scalar truth table and exact tower identities	Correctness and size of the charged multiplication primitives.
Streamed XL	Exact Hilbert coefficients, rows, columns, and operator products	Conditional low-state method; no unreproduced production profile is used in a headline.

14.1 Exact checks in the release

The release contains machine-readable ledgers and deterministic checkers. The main all-nine checker recomputes dimensions, separator choices, adaptive mixtures, target-sampling acceptance, and repair floors, while the separate field-circuit checker verifies the tower identities and all $19^2 = 361$ scalar products. The new Just Guess checker independently recomputes the expected-operation Boolean ledger, the capped-tree sensitivity, the quadratic-root truth table, the cross-channel second moment, and every displayed break-even threshold. The release does not claim that these consistency checks instantiate an official public key or run the production verifier; those are distinct fixed-key certificate tasks. The public leading-Jacobian preflight is deterministic on a fixed key; its failure invokes a complete-core fallback. The separate structural preflight is an exact condition for the displayed ell=4 routes and is not assigned a random success probability in the ledger.

14.2 What is not claimed

The release does not contain a production-size end-to-end forgery or a measured symbolic-homotopy implementation. In particular, the paper does not claim that κ_{hom} is small, that the direct parametrization fits in commodity memory, or that the unit-leading exponent is a complete NIST gate count. The AXN basis prices the multiplication schedule selected by the arithmetic theorem. Control, data movement, indexing, factorization, communication, and peak state remain outside that leading schedule and inside κ_{hom}. In the Just Guess branch, all field operations in the published expected schedule are converted to explicit Boolean-circuit costs and separate control/filtering reserves are displayed, but the candidate-generation regularity multiplier κ_{JG} remains to be measured or certified on a production key.

The low-state XL branch has the opposite profile: its streamed operator and state are explicit, but its production selected degree is not proved. A production border-basis transcript or a full-size solve would settle that remaining branch-specific question without changing the exact quotient or the selected-degree-free theorem.

14.3 Fixed-key end-to-end certificate format

A deterministic record for one public key consists of:

1. the public relation, output split, and all exact rank preflights;

2. the selected slice and either its exact projective rank histogram or a stronger unit-ideal certificate;

3. the fast-core leading-Jacobian witness, if the fast branch is taken;

4. the homotopy random coins, parametrization, and all candidate filters;

5. the reconstructed signature and the unmodified verifier transcript.

Such a record would remove the random-XOF key-density statement for that key and directly measure κ_{hom} and resident state.

15 A combined structural repair barrier

The feature-count floor and the fresh-square floor are independent necessary conditions and should be combined.

Proposition 15.1 (Universal fresh-square dimension bound)**.** *Let $A = \mathbb{F}_{q^d}$ and retain the public vinegar summand V' of base-field dimension $d(v-1)$. Suppose an exact affine reduction has M' outputs, K' quotient quadratics, and full affine rank $M' - K'$. Then*

$$\dim_{\mathbb{F}_q}(V' \cap \ker C_{R,v}) \geq d(v-1) - (M' - K').$$

If $M' \leq d(v-1)$, dimensions alone leave a complete ordinary K'-dimensional square slice. Eliminating this guarantee requires $M' > d(v-1)$. Under the coupling $M' = dx'$, this forces $x' \geq v$ and $M' \geq dv$.

Proof. Use

$$\dim(V' \cap \ker C_{R,v}) \geq \dim V' - \operatorname{rank} C_{R,v} = d(v-1) - (M' - K').$$

The right side is at least K' exactly when $M' \leq d(v-1)$. Negate and round to the next multiple of d. $\square$

Let $K_{\mathrm{ord}} = \min\{M, m_1 d^2\}$ be the old ordered-label cap. Retaining symmetric forms requires $m_1' \geq \lceil K_{\mathrm{ord}}/\binom{d+1}{2} \rceil$ and $M' \geq K_{\mathrm{ord}}$. Under $M' = dx'$ and $m_1' = \lceil x'/d \rceil$, all three necessities combine as follows.

Theorem 15.2 (Combined necessary output floor)**.** *Any redesign that retains symmetric public forms, the public vinegar count, the public-vinegar affine geometry above, full affine rank $M' - K'$, and the Version 2.3 output/base-form coupling, while both restoring K_{ord} and eliminating the guaranteed fresh square slice by dimensions, must satisfy*

$$x' \geq \max\left\{ v, \left\lceil \frac{K_{\mathrm{ord}}}{d} \right\rceil, d\left(\left\lceil \frac{K_{\mathrm{ord}}}{\binom{d+1}{2}} \right\rceil - 1 \right) + 1 \right\}, \qquad M' = dx'. \tag{64}$$

For the nine official rows:

Level	*parameters*	*current M*	*necessary M′*	*increase*
I	$(28, 5, 4, 4)$	80	116	45.00%
I	$(48, 16, 2, 2)$	64	96	50.00%
I	$(28, 4, 4, 5)$	80	116	45.00%
III	$(40, 7, 4, 4)$	112	180	60.71%
III	$(72, 24, 2, 2)$	96	144	50.00%
III	$(38, 5, 4, 5)$	100	152	52.00%
V	$(50, 9, 4, 4)$	144	228	58.33%
V	$(96, 32, 2, 2)$	128	192	50.00%
V	$(52, 6, 4, 6)$	144	228	58.33%

Thus every parameter-only repair retaining these hypotheses needs at least 45.00% *more verifier outputs, and the largest necessary increase is* 60.71%*, before adding any security margin. These are necessary, not sufficient, repaired parameters.*

Proof. The second and third terms in (64) are the two ordered-cap requirements under $m'_1 = \lceil x'/d \rceil$. The term v is Theorem 15.1. Their maximum is therefore necessary. Substitution gives the table. $\square$

15.1 A stronger chosen-message square-persistence barrier

The preceding floor assumes a full-rank affine lift and asks dimensions to remove a square slice from every target fiber. A zero-offset attacker has a second option: give up affine consistency for all targets, filter chosen-message targets through the quadratic image, and recover the entire common-column domain. Eliminating that fallback by dimensions alone is much more expensive.

Proposition 15.3 (Zero-offset chosen-message square persistence)**.** *Consider any symmetric redesign for which the zero-offset common-column relation with $R_0 = 1$ is defined. Let Q_R be its quadratic output matrix and put $\rho_Q = \operatorname{rank} Q_R$. Public row reduction gives an invertible output change under which the zero-offset verifier is equivalent to*

$$g_{\rho_Q}(u) = z \in \mathbb{F}_q^{\rho_Q}, \qquad t_{\text{con}} = 0 \in \mathbb{F}_q^{M'-\rho_Q}, \tag{65}$$

where g_{ρ_Q} consists of ρ_Q independent quadratic output combinations. For an exactly uniform target, the consistency event in (65) has probability $q^{-(M'-\rho_Q)}$ and, conditioned on it, z is uniform in $\mathbb{F}_q^{\rho_Q}$.

With zero offsets no reserved line is needed, so use the full public-vinegar space $V_{\text{vin}} \cong A^v$, of base-field dimension dv. If

$$\rho_Q \le dv, \tag{66}$$

then dimensions alone leave an ordinary ρ_Q-dimensional slice $L \subseteq V_{\text{vin}}$ on which the complete residual system is square: it has ρ_Q equations in ρ_Q variables. Consequently, ruling out this chosen-message square by dimensions alone requires

$$\rho_Q > dv. \tag{67}$$

Proof. With zero offsets, the substituted verifier is $Q_R z_{\text{sym}}(u) = t$ and contains no affine or constant term. Choose a row basis of Q_R and extend it to an invertible basis change on the M' output coordinates. The first ρ_Q transformed rows are the independent residual quadratics g_{ρ_Q}; the remaining rows vanish identically on the quadratic image and therefore impose the consistency equations in (65). An invertible linear transformation sends a uniform target to independent uniform residual and consistency coordinates, proving the distribution statement. Under (66), choose any ρ_Q-dimensional linear subspace of V_{vin} and restrict all ρ_Q residual equations to an affine coset of it. This gives the asserted complete square system. $\square$

Symmetry bounds the actual quotient rank by

$$\rho_Q \le \min\left\{M',\ m'_1\binom{d+1}{2}\right\}. \tag{68}$$

Hence making (67) even possible forces both terms on the right of (68) above dv.

Theorem 15.4 (Dimension-only floor against the zero-offset fallback)**.** *Retain symmetry, the public vinegar count, and the Version 2.3 coupling*

$$M' = dx', \qquad m'_1 = \left\lceil \frac{x'}{d} \right\rceil.$$

Put

$$C_d = \binom{d+1}{2}, \qquad m_{\mathrm{sq}} = \left\lfloor \frac{dv}{C_d} \right\rfloor + 1.$$

A necessary condition for eliminating the zero-offset chosen-message square by dimensions alone is

$$x' \ge \max\left\{v+1,\ d(m_{\mathrm{sq}}-1)+1\right\}, \qquad M' = dx'. \tag{69}$$

If the redesign also restores the nominal ordered-label cap K_{ord}, the right side of (69) must additionally be maximized with

$$\left\lceil \frac{K_{\mathrm{ord}}}{d} \right\rceil, \qquad d\left(\left\lceil \frac{K_{\mathrm{ord}}}{C_d} \right\rceil - 1\right) + 1.$$

For the nine official rows, the zero-offset condition already dominates the ordered-cap terms:

Level	*parameters*	*current M*	*dimension-only M'*	*increase*
I	$(28, 5, 4, 4)$	80	180	125.00%
I	$(48, 16, 2, 2)$	64	130	103.125%
I	$(28, 4, 4, 5)$	80	180	125.00%
III	$(40, 7, 4, 4)$	112	260	132.14%
III	$(72, 24, 2, 2)$	96	194	102.08%
III	$(38, 5, 4, 5)$	100	244	144.00%
V	$(50, 9, 4, 4)$	144	324	125.00%
V	$(96, 32, 2, 2)$	128	258	101.5625%
V	$(52, 6, 4, 6)$	144	324	125.00%

Thus eliminating this fallback by dimensions alone requires 101.5625%–144.00% *more outputs across the published rows.*

Proof. Condition (67) and the two caps in (68) require $dx' > dv$ and $m'_1 C_d > dv$. The first inequality is equivalent to $x' \ge v+1$. The second requires $m'_1 \ge m_{\mathrm{sq}}$, and the least integer x' with $\lceil x'/d \rceil \ge m_{\mathrm{sq}}$ is $d(m_{\mathrm{sq}}-1)+1$. This proves (69). The two extra ordered-cap terms are the same feature and output necessities used in Theorem 15.2. Direct substitution gives the table. □

How to interpret the stronger floor. This is deliberately a dimension-only theorem, not a claim that every secure repair must literally double its outputs. A redesign below the table can instead leave the square system present and try to obtain security from the chosen-message consistency factor $q^{M'-\rho_Q}$, from the cost of solving the square system, or from a changed target interface. It must then publish and price those ingredients explicitly. The theorem gives a sharp trichotomy: break the symmetric/affine architecture, nearly double the outputs to remove the fallback by dimensions, or retain the fallback and justify its complete chosen-message cost.

16 Conclusion

The $q = 19$ Version 2.3 construction loses quadratic directions for every symmetric public key. Ordered power-pair labels are not ordered polynomials: swapping the labels leaves the scalar quadratic unchanged before the public outer map. Exact output splitting turns that identity into complete forgery systems without discarding a verifier equation, and the block-symmetry rejection rule does not remove them.

Every published row leaves a complete square residual system. Finite-field homotopy gives selected-degree-free, fixed-core-free recovery, but its all-nine numerical comparison is a unit-leading break-even ledger with an explicit, unmeasured κ_{hom}. The correct adaptive thresholds are 10.812, 23.205, and 38.156 bits of allowable overhead at Levels I, III, and V; the paper does not identify those values with complete classical-gate bounds.

The new Just Guess route addresses the most serious numerical objection more concretely. For the Level-I $\ell = 2$ row it replaces symbolic homotopy by a Boolean-ledger instantiation of the published expected schedule of guesses, quadratic roots, linear solves, quotient checks, and verifier calls. The expected trial ledger is below $2^{132.047}$; an exact conditional-channel atom calculation and a cross-channel second moment yield the success-adjusted expression $2^{135.426}\kappa_{\mathrm{JG}}$. Thus the Level-I comparison is below 2^{143} when $\log_2 \kappa_{\mathrm{JG}} < 7.575$. This is a substantially narrower and fixed-key-testable premise than κ_{hom}, but it is still a premise, not an unconditional complete-gate break.

A repair retaining the architecture cannot be a small rejection patch. Restoring the former feature cap and eliminating the guaranteed full-affine square already requires 45.00%–60.71% more verifier outputs under the current coupling. Eliminating the zero-offset chosen-message square by dimensions alone requires 101.5625%–144.00% more. A smaller redesign must retain and explicitly price the square system and target filter, change the target interface, or abandon the symmetric/affine architecture.

A Exact fixed-key spectrum certificates

The random-XOF theorem is not required for a statement about a concrete key. This appendix gives a finite exact certificate for the aggregate spectrum.

Choose a basis $y_0, \ldots, y_{h-1}$ of a fixed slice L and write

$$T(Y) = \sum_{i=0}^{h-1} Y_i T_{y_i} \in \mathbb{F}_q[Y_0, \ldots, Y_{h-1}]^{K \times N}.$$

For $0 \le j < h$, use the disjoint normalized projective chart $Y_0 = \cdots = Y_{j-1} = 0$, $Y_j = 1$, and put $R_j = \mathbb{F}_q[Y_{j+1}, \ldots, Y_{h-1}]$. Let $T^{(j)}$ be the resulting matrix. For $t \ge 0$, define

$$I_{j,t} = \left\langle Y_i^q - Y_i \ (i > j), \quad \text{all } (t+1) \times (t+1) \text{ minors of } T^{(j)} \right\rangle \subseteq R_j. \tag{70}$$

Put $n_{j,\le t} = \dim_{\mathbb{F}_q}(R_j/I_{j,t})$, $n_{j,\le -1} = 0$, and

$$N_t = \sum_{j=0}^{h-1} (n_{j,\le t} - n_{j,\le t-1}). \tag{71}$$

Proposition A.1 (Projective rank-histogram certificate)**.** *The integer N_t is exactly the number of projective directions $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$ for which* $\operatorname{rank} T_y = t$*. Consequently*

$$\bar{\eta}_L^U = (q-1) \sum_t N_t q^{-t}. \tag{72}$$

Reduced Gröbner bases with their standard monomial sets, or polynomial reduction transcripts proving the unit-ideal and quotient-dimension claims, form independently checkable fixed-key certificates.

Proof. The normalized charts contain exactly one representative of every projective $\mathbb{F}_q$-direction. The field equations identify

$$R_j/\langle Y_i^q - Y_i : i > j \rangle \cong \prod_{a \in \mathbb{F}_q^{h-j-1}} \mathbb{F}_q$$

by evaluation. Every ideal in this product of fields is radical, and its quotient dimension is the number of coordinates on which all generators vanish. The minors in (70) vanish exactly when $\operatorname{rank} T^{(j)}(a) \le t$. Thus $n_{j,\le t}$ counts rank-at-most-t points in chart j, and differencing gives (71). Every projective direction has $q-1$ nonzero scalar representatives and rank is invariant under base-field scaling, proving (72). □

A stronger certificate that $\operatorname{rank} T_y \ge r+1$ for every nonzero y consists of unit-ideal transcripts for all $I_{j,r}$. The complete histogram is more flexible: a small exceptional low-rank locus can be charged exactly instead of controlling the whole pencil by its minimum rank.

B Dimension formulas and exact numerical rows

For a Version 2.3 row,

$$n = v + o, \quad m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad N_0 = n\ell, \quad K = m_1 \binom{\ell+1}{2}.$$

For a full affine lift with $\operatorname{rank} C_{R,v} = M - K$,

$$\dim \ker C_{R,v} = N_0 - M + K.$$

For a structured lift whose complete offset orbit has rank $m_1 \ell^2$,

$$\dim_A H_{R,v}^F = n - m_1 \ell.$$

The direct ordinary fresh-slice lower bound for $\ell = 4$ is

$$\dim(V' \cap \ker C_{R,v}) \ge 4(v-1) - (M-K).$$

Table 5: Audited exact unit-leading exponents underlying the rounded main text, conditional on the exact public structural preflight for the $\ell = 4$ rows. "Direct" solves the complete square; "adaptive" is the conditional mixture of an exact fast-core preflight and the complete-square fallback.

Level	Parameters	Direct	Adaptive	Adaptive headroom
I	$(28, 5, 4, 4)$	142.606960	128.246278	14.753722
I	$(48, 16, 2, 2)$	132.970670	132.188072	10.811928
I	$(28, 4, 4, 5)$	142.606374	128.245747	14.754253
III	$(40, 7, 4, 4)$	185.450879	171.616678	35.383322
III	$(72, 24, 2, 2)$	183.794625	183.794625	23.205375
III	$(38, 5, 4, 5)$	185.450045	171.615857	35.384143
V	$(50, 9, 4, 4)$	227.560406	214.558321	57.441679
V	$(96, 32, 2, 2)$	233.843987	233.843987	38.156013
V	$(52, 6, 4, 6)$	227.559359	214.557277	57.442723

All values charge the 150, 692, and 2628-gate multiplication circuits as appropriate and use the conservative B^2 forbidden-vector separator. For $\ell = 4$, the Jacobian-preflight density uses the projective right-kernel union bound, not an independence claim after a biased linear transform.

C Arithmetic-circuit certificate details

The scalar multiplier consumes two little-endian five-bit canonical integers in $\{0, \dots, 18\}$ and returns the canonical five-bit representation of their product modulo 19. Its 150 gates are two-input AND, XOR, and XNOR gates with free constants, wires, fan-out, and structural sharing. The released netlist is tested on all 361 valid input pairs. Its SHA-256 digest is

`e1708a32c1475ad14295c2b34fed976fb09577b127810d29bf237db18aabc131`

for the audited balanced-netlist representation.

The tower recurrences are

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2 + 1), \qquad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2 - (1 + u)).$$

Three-product Karatsuba at each level, together with the exact linear transforms stated in Theorem 7.1, gives complete multiplication counts 692 and 2628. The artifact independently verifies the tower identities, the basis determinant, and random products against direct polynomial arithmetic.

These circuits normalize only the pointwise multiplication schedule. They are not lower bounds and do not price the whole solver. Additions, inversions, factorization, fixed extension transforms, filtering, control, indexing, communication, and memory remain in κ_{hom}.

D Reproduction and package contents

The source package is self-contained. Its root contains the complete paper source, bibliography, compiled PDF, build instructions, and hashes. The `artifact/` directory contains:

- a standalone generator and independently coded checker for the corrected all-nine conditional homotopy ledger;
- the complete 150-gate scalar multiplier netlist and tower checks;
- the Just Guess expected-operation and capped ledgers, the quadratic-root check, and the exact conditional Frobenius-channel atom enumeration; and

- a wrapper checker that runs both the all-nine and referee-revision checks.

A clean reproduction is:

```
sha256sum -c SHA256SUMS
make regenerate
make verify
make
```

The all-nine generator assigns no random probability to the fixed outer map: its $\ell = 4$ ledgers condition on the exact public structural preflight. The checker is deterministic, requires Python 3.10 or newer, and does not require network access. The paper build uses only the files in the package. The supplied `SHA256SUMS` records every distributed file.

References

[1] National Institute of Standards and Technology. *Nine Candidates Advance to the Third Round of the Additional Digital Signatures for the PQC Standardization Process*. NIST news release. May 2026. URL: https://csrc.nist.gov/News/2026/nist-advances-9-candidates-to-the-3rd-round-of-pqc.

[2] Gorjan Alagic, Maxime Bros, Pierre Ciadoux, Quynh Dang, Thinh Hung Dang, John Kelsey, Jacob Lichtinger, Yi-Kai Liu, Carl Miller, Dustin Moody, Rene Peralta, Ray Perlner, Angela Robinson, Hamilton Silberg, Daniel Smith-Tone, and Noah Waller. *Status Report on the Second Round of the Additional Digital Signature Schemes for the NIST Post-Quantum Cryptography Standardization Process*. Tech. rep. NIST IR 8610. National Institute of Standards and Technology, May 2026. DOI: 10.6028/NIST.IR.8610. URL: https://doi.org/10.6028/NIST.IR.8610.

[3] Maxime Bros, Thai Hung Le, Jacob Lichtinger, Brice Minaud, Ray A. Perlner, Daniel Smith-Tone, and Cristian Valenzuela. *Exploiting SNOVA's Structure in the Wedge Product Attack*. Cryptology ePrint Archive, Paper 2026/237. 2026. URL: https://eprint.iacr.org/2026/237.

[4] Po-En Tseng, Lih-Chung Wang, Peigen Li, and Yen-Liang Kuan. *Investigating the Wedge Map on SNOVA*. Cryptology ePrint Archive, Paper 2026/260. 2026. URL: https://eprint.iacr.org/2026/260.

[5] SNOVA Team. *SNOVA Version 2.3 Reference Implementation and Documentation*. Git commit 9da14981336ede257c41ef53cc069989051e8181. Immutable commit cited for parameter generation, symmetric public forms, rejection rules, and fixed SNOVA_ABQ behavior; accessed 25 July 2026. Apr. 2026. URL: https://github.com/PQCLAB-SNOVA/SNOVA/tree/9da14981336ede257c41ef53cc069989051e8181.

[6] Ward Beullens. "Improved Cryptanalysis of SNOVA". In: *Advances in Cryptology – EUROCRYPT 2025*. Vol. 15606. Lecture Notes in Computer Science. Cryptology ePrint Archive, Paper 2024/1297. Springer, 2025, pp. 277–293. DOI: 10.1007/978-3-031-91095-1_10. URL: https://eprint.iacr.org/2024/1297.

[7] National Institute of Standards and Technology. *Post-Quantum Cryptography: Security Evaluation Criteria*. NIST PQC project page. Living project page; classical-gate reference exponents 143, 207, and 272. June 16, 2026. URL: https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/evaluation-criteria/security-(evaluation-criteria) (visited on 07/26/2026).

[8] Daniel Cabarcas, Peigen Li, Javier A. Verbel, and Ricardo Villanueva-Polanco. "Improved Attacks for SNOVA by Exploiting Stability under a Group Action". In: *Advances in Cryptology – CRYPTO 2025*. Vol. 16005. Lecture Notes in Computer Science. Cryptology ePrint Archive, Paper 2024/1770. Springer, 2025, pp. 358–389. DOI: 10.1007/978-3-032-01887-8_12. URL: https://eprint.iacr.org/2024/1770.

[9] Hiroki Furue, Yasuhiko Ikematsu, Shuhei Nakamura, and Rika Akiyama. "Improved Cryptanalysis of SNOVA by Solving Multi-Homogeneous Systems via Matrix Transformations". In: *Advances in Cryptology – ASIACRYPT 2025*. Vol. 16248. Lecture Notes in Computer Science. Springer, 2025, pp. 405–435. DOI: 10.1007/978-981-95-5113-2_13.

[10] Jean-Charles Faugère, Patrizia Gianni, Daniel Lazard, and Teo Mora. "Efficient Computation of Zero-Dimensional Gröbner Bases by Change of Ordering". In: *Journal of Symbolic Computation* 16.4 (1993), pp. 329–344. DOI: 10.1006/jsco.1993.1051.

[11] Achim Kehrein and Martin Kreuzer. "Characterizations of Border Bases". In: *Journal of Pure and Applied Algebra* 196.2–3 (2005), pp. 251–270. DOI: 10.1016/j.jpaa.2004.08.028.

[12] Mohab Safey El Din and Éric Schost. "Bit Complexity for Multi-Homogeneous Polynomial System Solving: Application to Polynomial Minimization". In: *Journal of Symbolic Computation* 87 (2018), pp. 176–206. DOI: 10.1016/j.jsc.2017.08.001. arXiv: 1605.07433.

[13] Lih-Chung Wang, Chun-Yen Chou, Jintai Ding, Yen-Liang Kuan, Jan Adriaan Leegwater, Ming-Siou Li, Bo-Shu Tseng, Po-En Tseng, and Chia-Chun Wang. *On the Security of Round 2 SNOVA*. Sixth NIST Post-Quantum Cryptography Standardization Conference presentation. Presented 26 September 2025; accessed 27 July 2026. Sept. 2025. URL: https://csrc.nist.gov/presentations/2025/on-the-security-of-round-2-snova.

[14] Yasuhiko Ikematsu and Rika Akiyama. "Revisiting the Security Analysis of SNOVA". In: *Proceedings of the 11th ACM Asia Public-Key Cryptography Workshop*. Cryptology ePrint Archive, Paper 2024/096. ACM, 2024, pp. 54–61. DOI: 10.1145/3659467.3659900. URL: https://eprint.iacr.org/2024/096.

[15] Peigen Li and Jintai Ding. "Cryptanalysis of the SNOVA Signature Scheme". In: *Post-Quantum Cryptography – PQCrypto 2024*. Vol. 14772. Lecture Notes in Computer Science. Cryptology ePrint Archive, Paper 2024/110. Springer, 2024, pp. 79–91. DOI: 10.1007/978-3-031-62746-0_4. URL: https://eprint.iacr.org/2024/110.

[16] Shuhei Nakamura, Yusuke Tani, and Hiroki Furue. "Lifting Approach against the SNOVA Scheme". In: *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences* E108.A.10 (2025), pp. 1373–1381. DOI: 10.1587/transfun.2024EAP1124.

[17] Lars Ran. "Wedges, Oil, and Vinegar: An Analysis of UOV in the Exterior Algebra". In: *Advances in Cryptology – EUROCRYPT 2026*. Vol. 16544. Lecture Notes in Computer Science. Springer, 2026, pp. 363–388. DOI: 10.1007/978-3-032-25327-9_13.

[18] Yi Jin, Yuansheng Pan, Xiaoou He, Boru Gong, and Jintai Ding. "Security Analysis on UOV Families with Odd Characteristics: Using Symmetric Algebra". In: *Public-Key Cryptography – PKC 2026*. Vol. 16551. Lecture Notes in Computer Science. Cryptology ePrint Archive, Paper 2025/1137. Springer, 2026, pp. 428–454. DOI: 10.1007/978-3-032-26731-3_14. URL: https://eprint.iacr.org/2025/1137.

[19] Yasufumi Hashimoto. "An Improvement of Algorithms to Solve Under-Defined Systems of Multivariate Quadratic Equations". In: *JSIAM Letters* 15 (2023), pp. 53–56. DOI: 10.14495/jsiaml.15.53.

[20] Alexander May, Massimo Ostuzzi, and Henrik Ressler. "Just Guess: Improved (Quantum) Algorithm for the Underdetermined MQ Problem". In: *Advances in Cryptology – EUROCRYPT 2026*. Vol. 16544. Lecture Notes in Computer Science. Cryptology ePrint Archive, Paper 2025/1788. Springer, 2026, pp. 334–362. DOI: 10.1007/978-3-032-25327-9_12. URL: https://eprint.iacr.org/2025/1788.

[21] Hideki Asanuma, Yilong Chen, Hiroki Furue, Kosuke Sakata, and Tsuyoshi Takagi. *Improved Complexity Estimates for Underdetermined MQ Systems via Generalized Variable Partitioning*. Cryptology ePrint Archive, Paper 2026/1054. 2026. URL: https://eprint.iacr.org/2026/1054.

[22] Kosuke Sakata and Tsuyoshi Takagi. "An Efficient Variant of F4 Algorithm for Solving MQ Problem". In: *IACR Transactions on Cryptographic Hardware and Embedded Systems* 2026.3 (2026), pp. 1284–1309. DOI: 10.46586/tches.v2026.i3.1284-1309.